

Kaway-Kawayan



Bamboo
Music
Education
Lessons
for K-12

Volume 1

Elementary

Kaway-Kawayan
Bamboo Music Education Lessons for K-12

Volume 1
Elementary Set

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TABLE OF CONTENTS

INTRODUCTION	i-ii
MODULE 1: BAMBOO ORGAN OF LAS PIÑAS (GRADE 4)	1-4
OBJECTIVES.....	1
CONTENT.....	1
LEARNING RESOURCES.....	1-2
PROCEDURE	2-3
TEACHER'S REMARKS.....	3
REFLECTION.....	4
MODULE 2: BAMBOO UKULELE OF CEBU (GRADE 6).....	5-8
OBJECTIVES.....	5
CONTENT.....	5
LEARNING RESOURCES.....	5-6
PROCEDURE	6-7
TEACHER'S REMARKS.....	7
REFLECTION.....	8
MODULE 3: HIMIG KAWAYAN OF QUEZON CITY,	9-14
LAGUNA, STA. MESA (GRADE 5)	
OBJECTIVES.....	9
CONTENT.....	9
LEARNING RESOURCES.....	9-10
PROCEDURE	10-13
TEACHER'S REMARKS.....	13
REFLECTION.....	14
MODULE 4: KARATONG FESTIVAL OF DULAG, LEYTE (GRADE 5).....	15-18
OBJECTIVES.....	15
CONTENT.....	16
LEARNING RESOURCES.....	16
PROCEDURE	16-18
TEACHER'S REMARKS.....	18
REFLECTION.....	18

MODULE 5: MUSIKONG BUMBONG OF ANGONO, RIZAL (GRADE 5)	19-24
OBJECTIVES.....	19
CONTENT.....	19
LEARNING RESOURCES.....	20
PROCEDURE	21-23
TEACHER'S REMARKS.....	24
REFLECTION.....	24
 MODULE 6: MUSIKONG BUMBONG OF BULACAN (GRADE 5)	 25-29
OBJECTIVES.....	25
CONTENT.....	25
LEARNING RESOURCES.....	25-26
PROCEDURE	26-28
TEACHER'S REMARKS.....	29
REFLECTION.....	29
 MODULE 7: TULTUGAN MUSIC OF ILOILO (GRADE 5)	 30-33
OBJECTIVES	30
CONTENT.....	30
LEARNING RESOURCES	31
PROCEDURE	31-32
TEACHER'S REMARKS.....	33
REFLECTION.....	33
 APPENDICES.....	 34-80
APPENDIX A: ETHNOGRAPHIES	34
APPENDIX B: PHOTOS	58
APPENDIX C: URL LINKS.....	67
APPENDIX D: MUSIC SCORES.....	69

INTRODUCTION

“Sa Filipinas, ang kawayan ay malaking bahagi ng kultura dahil kaugnay ito ng iba’t ibang tradisyon, pagdiriwang, at paniniwala. Halimbawa, sa panitikan, ayon sa alamat, ang unang lalaki at unang babae ay nagmula sa isang pirasong kawayan. Sa sayaw, ang bantog na tinikling, singkil at subli ay isinasayaw gamit ang kawayan. Sa musika, ginagamit din ang kawayan sa paggawa ng iba’t ibang klase ng instrumentong pangmusika. Maging ang mga katutubong laro gaya ng luksong kawayan at palosebo ay gumagamit ng damong tropikong ito.”

(In the Philippines, bamboo plays a big part in culture because of its varied roles in tradition, celebration, and folk beliefs. According to Philippine folk literature, the first man and woman came from a bamboo pole split into two. Philippine dances such as the tinikling, singkil, and subli use bamboo accompaniment. Bamboo is also used in various types of Philippine musical instruments. Philippine games like luksong kawayan and palosebo use the tropical grass.)

(Source: <https://philippineculturaleducation.com.ph/kawayan/>)

In the distant past, our ancestors told a story about how the first Filipinos were born from the bamboo. The story goes that a bird was exploring an island when it heard a voice coming from a bamboo pole. Curious, the bird tapped on the bamboo. The bamboo cracked open and there emerged from it a man and a woman. The bird carried the two on its back toward an island where life as we know it began. At present, bamboo continues to play an important part in our lives. We use it as ingredient and tools in our cuisine, as medium in our arts and music, as building materials in our furniture and architecture, and as part of our decorative landscaping and flood and erosion control systems.

Like the first man and woman in the folktale, **Kaway-Kawayan** is a child of the bamboo, born from the labors of the Department of Science and Technology Forest Products Research and Development Institute (DOST-FPRDI), the UP Center for Ethnomusicology (UPCE), the Philippine Normal University (PNU), and the Philippine Society for Music Education (PSME). Together, these institutions and organizations worked on the Bamboo Music Instruments Innovation Research and Development Program, which is composed of projects seeking to develop the bamboo music industry in the Philippines through ethnographic and scientific documentation of folk and commercial bamboo makers, treatment of commonly damaged bamboo musical instruments, sound processing techniques and analysis, and bamboo processing and training center.

Kaway-Kawayan is a collection of music lessons on bamboo music integrating the content and performance standards of the Philippine K to 12 music curriculum. The lessons contained herein are a product of the collaboration of researchers from UPCE and writers from PNU and PSME. The ethnographic documentation of UPCE yielded significant records of bamboo music traditions in the Philippines, updating the research done by National Artist Jose Maceda in the 1970s until the 1980s. These evolving folk knowledge, skills, and practices documented in the ethnography of selected areas are imparted through the lessons written by the writers who were present during the fieldwork done by the ethnographers documenting the bamboo music traditions in situ or via online.

The first volume of **Kaway-Kawayan** focuses on the following Philippine bamboo music traditions and instrument makers: (1) Bamboo Organ of Las Piñas, (2) Bamboo Ukulele of Cebu, (3) Himig Kawayan of Quezon City, Laguna, Sta. Mesa; (4) Karatong Festival of Dulag, Leyte; (5) Musikong Bumbong of Angono, Rizal; (6) Musikong Bumbong of Bulacan; and (7) Tultugan Music of Iloilo (the Bamboo Rondalla of Dipolog is included only in the High School set). The lessons combine the ethnographic information contained in the bamboo music documentation with a sequential plan of musical experiences that are age-appropriate, culturally sensitive, and integrative of the Filipino children or youth's evolving musicking.

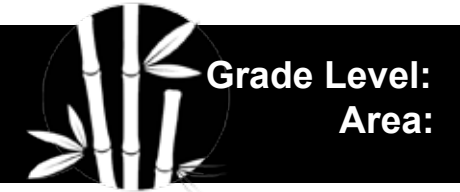
This volume has two sets: elementary (grades one through six) and junior high school (grades seven through ten). Content, standards, and competencies are derived from the DepEd K to 12 music curriculum. The music strategies reflect active listening, performing vocally and instrumentally alone or with others, evaluating one's music performance, and integrating bamboo music performances with related arts like speech choir, dance, or theatre. Materials used in the lessons were carefully selected from the ethnographic documentation and online video materials with accompanying links for media access. Reflections after each lesson enable the teacher to look deeper into the various lesson components and remediation activities when needed. Enrichment activities are also provided for learners who want to know more about bamboo traditions in general and bamboo instrument makers from the research sites in particular. When applicable, interdisciplinary activities for mathematics, social studies, physical and social sciences are included in some of the lessons.

Finally, the word "**kaway-kawayan**" is also a wordplay that means "bamboo waving." This wordplay is an apt metaphor for the swaying bamboo as it bends against adversity, never breaking but always adapting. We hope that through the lessons contained in Kaway-Kawayan, teachers, students, and bamboo enthusiasts can derive, interpret, and adapt educative experiences to keep the Philippine music like the bamboo – with roots firmly attached to our traditions and culture, but at the same time continuously growing and shooting upwards to face the needs and challenges of the present and the future.

Jocelyn Timbol-Guadalupe

BAMBOO ORGAN OF LAS PIÑAS

Prepared by: ANNA PATRICIA RODRIGUEZ-CARRANZA



Grade Level:
Area:

4
Music

I. OBJECTIVES

A. Content Standards

Demonstrate understanding of variations of sound in music (lightness and heaviness) as applied to vocal and instrumental music

B. Performance Standards

Participate actively in a group performance to demonstrate different vocal and instrumental sounds

C. Learning Competencies/Objectives

1. Recognize that the bamboo organ can recreate different instrumental sounds
2. Classify the bamboo organ sound patches as batalla (trumpet), clarinet, flauto (flute/recorder-like), pajaritos (bamboo bird whistle with water), tambor (percussive)

II. CONTENT

Timbre - variations of sound

A. Voicing the organ

B. Instrumental sounds of the bamboo organ stops

III. LEARNING RESOURCES

A. References

1. Print Materials

BMI ethnography of Las Piñas Bamboo Organ

2. Audio-Video Materials

a. BMI video of the Bamboo Organ demonstration by Armando Salarza:
<http://tiny.cc/BMI-Bamboo-Organ>

b. BMI Pictures of the Bamboo Organ pipes and stops

B. Other Learning Materials

1. Small handheld percussion instruments
2. Clean plastic/glass bottles (in various sizes filled with water, rocks, small beads, or sand, etc.)

IV. PROCEDURE

A. Reviewing the previous lesson or presenting the new lesson

1. Ask the students who among them have visited a place of mass/worship. Facilitate a discussion to what types of music they might hear/have heard in a church/ mosque/ chapel. Lead the discussion to the St. Joseph Parish in Las Piñas, wherein a special kind of instrument is used not only in mass, but in concerts and other performances as well.
2. Show pictures of the Bamboo Organ, with a few clips (audio, but preferably video) of the demonstration by organist Prof. Armando Salarza.

B. Establishing the purpose of the lesson

Tell the students that they will be identifying the different sounds the bamboo organ can recreate, which they will relate with the instrument groupings in the orchestra. After this, the students will try to recreate the said different kinds of sounds using vocal sounds and clean plastic/glass bottles.

C. Presenting the new lesson

Listen and Imitate: Have the students listen to the isolated sounds of each stop. Name each sound patch with the proper label.

D. Discussing new concepts and practicing new skills # 1

1. Have the students imitate the sound patch by exploring their vocal sounds. Tell them that just like the bamboo organ's stops, humans can modify their voices to produce different timbres.
2. After the students have explored their vocal sounds, ask them to apply the different vocal timbres that imitate the bamboo organ stops in singing a major scale (skill learned in the second quarter).
3. A video of the interview with organ builder Cealwyn Tagle on voicing the organ can be shown.

E. Discussing new concepts and practicing new skills # 2

1. Divide the class into groups. Assign each group one type of particular filler (water, rocks, beads, pebbles, sand, etc.) to fill up their bottles halfway.
2. Let the students blow the bottle to produce a sound. Let the students compare the sounds they've produced. (The bottle is blown like the saggeypo. Also, be sure to have one bottle per student for sanitary health reasons).

F. Developing mastery (leads to formative assessment)

1. Create a sound piece utilizing vocal sounds, bottles, and some handheld percussions. Guide the students in writing their sound piece using any system of notation that will work for them (symbols, pictures, lines, Western notation, numbers, etc.).
2. Let the students perform the piece they created. Record the full run and let the students listen to it with the score they made.

G. Finding practical applications of concept and skills in daily living

Prompt: How can I create different kinds of sounds (timbre) using only one object?

H. Making generalization and abstraction about the lesson

Different sounds can result from the same material through different manipulations. For example, the bamboo organ can produce different kinds of sounds because of the different stops that it has.

I. Additional activities for application or remediation

Students who find it hard to blow the bottles can instead use a plastic recorder, which they can play in different safe ways.

V. TEACHER'S REMARKS

VI. REFLECTION

- A. How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
- B. Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
- C. Which of my teaching strategies worked well? Why did this work?
- D. What difficulties can my principal or supervisor help me solve?
- E. What innovation or localized materials did I use/discover that I wish to share with other teachers?



Grade Level:
Area:

BAMBOO UKULELE OF CEBU

Prepared by: DOLORES TEXON-ANDRES

6
Music

I. OBJECTIVES

A. Content Standards

1. Demonstrate understanding of the concept of timbre through recognizing instruments aurally and visually
2. Demonstrate understanding of the concepts of harmony through the intervals that constitute the primary chords of major and minor scales in music

B. Performance Standards

1. Participate in a group performance to demonstrate different vocal and instrumental sounds
2. Perform a vocal or instrumental ensemble using the following major triads (I, IV, V)

C. Learning Competencies/Objectives

1. Identify aurally and visually a bamboo ukulele
2. Play the basic chords (I, IV, V) on ukulele
3. Use ukulele to accompany singing of folk songs or popular songs

II. CONTENT

1. Timbre - variations of sound in instrumental music, melodic and harmonic sound of bamboo ukulele
2. Harmony - basic chords (I, IV, V)

III. LEARNING RESOURCES

A. References

1. Print Materials
 - a. BMI ethnography of Cebu Bamboo Ukulele
 - b. Photos of Cebu's Huni Ukuleles
 - c. Website of Cebu's Huni Ukuleles: <https://huniukes.weebly.com/>

2. Audio-Video Materials

- a. Huni Soprano Ukulele
<http://tiny.cc/YT-HuniUkuleles-Soprano>
- b. Huni Tenor Ukulele
<http://tiny.cc/YT-HuniUkuleles-Tenor>

3. Music Scores

“Tao, Tao Po,” Tagalog folk song

B. Other Learning Materials

1. Ukulele instruments
2. Basic ukulele chord chart

IV. PROCEDURE

A. Reviewing the previous lesson or presenting the new lesson

Show pictures of bamboo handicrafts like bamboo candleholders, bilao, basket, furniture, tray, and bamboo ukulele. Have a discussion on the qualities of bamboo as a tree, product, and instrument.

B. Establishing the purpose of the lesson

Show pictures Huni Ukuleles bamboo ukulele and let the pupils describe its design. Narrate the craftsmanship in the process of making ukulele instruments.

C. Presenting examples/instances of the new lesson

Watch the video on Huni Soprano Ukulele and Tenor Ukulele. Let the pupils describe the sound quality of the soprano and tenor ukulele.

D. Discussing new concepts and practicing new skills # 1

1. Explain the three basic major chords. Show the chord chart of C, F, and G. Play each chord four times(4x) in quarter note rhythm.
2. Introduce the term “strophic” as the term that describes the form of the song where the tone and rhythm are repeated in each verse of the song.

E. Discussing new concepts and practicing new skills # 2

1. Sing “Tao, Tao po.”
2. Practice the chord progression of the song.
3. Use this strum pattern:
D D DU D (D-down U-Up)
2/4 C C - C C - C C - G G - G G - G G - G G - C C
C C - C C - C C - F F - F F - C C - G G - C C
4. Divide the class into groups. One group will sing while the others will play the accompaniment on the ukulele.

F. Developing mastery (leads to formative assessment)

1. Play the chords of “You Are My Sunshine.”
2. Use the strum pattern: D D D DU

4/4 Intro C C C C -
C C C C - C C C C - F F F F - C C C C
F F F F - C C C C - G G G G - C C C C

3. Sing and play “You Are My Sunshine”

G. Finding practical applications of concept and skills in daily living

Prompt 1: In the lesson, what did you like most? Singing or playing ukulele?
Were you able to sing and play at the same time?

Prompt 2: In our everyday life, there are tasks that you can do alone and there are tasks when we need help or assistance from others.

H. Making generalization and abstraction about the lesson

Let the pupils name and explain the hierarchy of relationships of basic chords.

I. Additional activities for application or remediation

Watch a medley of Cebuano folk songs accompanied by the bamboo ukulele.
Sing and play some of the familiar songs together with the video

V. TEACHER’S REMARKS

VI. REFLECTION

- A. How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
- B. Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
- C. Which of my teaching strategies worked well? Why did this work?
- D. What difficulties can my principal or supervisor help me solve?
- E. What innovation or localized materials did I use/discover which I wish to share with other teachers?

HIMIG KAWAYAN OF QUEZON CITY, LAGUNA, STA. MESA

Prepared by: PACITA M. NARZO



Grade Level: 5
Area: Music

I. OBJECTIVES

A. Content Standards

Demonstrate understanding of variations of sound density in music (lightness and heaviness) as applied to vocal and instrumental music

B. Performance Standards

Participate in a group performance to demonstrate different vocal and instrumental sounds

C. Learning Competencies/Objectives

1. Identify aurally and visually different instruments in Himig Kawayan/Pangkat Kawayan
2. Participate actively in instrumental ensemble

II. CONTENT

Instrumental music of Pangkat Himig Kawayan

1. Angklung
2. Marimba
3. Bumbong

III. LEARNING RESOURCES

A. Materials

1. Print Materials

BMI ethnography on angklung, marimba and bumbong:

- a. BMI ethnography of Quezon City's Pangkat Kawayan
- b. BMI ethnography of Laguna's Kawayan Music Ensemble
- c. BMI ethnography of Sta. Mesa's Banda Kawayan

2. Audio-Video Materials

- a. Angklung Videos
 - Angklung – Himig Kawayan tutorial
 - Indonesian Angklung
 - Angklung Performance
 - b. Marimba Videos
 - Marimba – Himig Kawayan tutorial
 - Philippine Music: Dipolog Marimba Nais Ko
 - Banda Kawayan Plays Pinoy Music
 - c. Bumbong Videos
 - Himig Kawayan tutorial
 - d. Pangkat Kawayan Videos
 - e. Musikawayan /D' Original Musikong Bumbong 1896 Videos
 - f. PUP Banda Kawayan/Banda Kawayan Pilipinas Videos
 - g. Bocaue Central School Musikawayan Videos
 - h. Musikawayan ng DepEd, Bocaue Videos
- ## 3. Music Scores
- a. “Manang Biday,” arranged by Genaro Cesar Edilo Jr.
 - b. “Pambansang Awit,” arranged by Genaro Cesar Edilo Jr.

B. Other Learning Materials

1. Piano Chords Tutorial Videos
2. Basic Articulation Videos

IV. PROCEDURE

A. Reviewing the previous lesson or presenting the new lesson

1. Discuss *legato* and *staccato* articulation through singing and listening to musical excerpts.
2. Review lessons on chord triads.

B. Establishing the purpose of the lesson

Tell the students that through the discussion on the Himig Kawayan ensemble, they will be able to:

1. Identify visually and aurally the basic instruments of the ensemble
2. Explain the functions of the instruments and describe basically how the instruments are played

3. Play with some of the instruments of the ensemble
4. Identify some of the founders/ manufacturers of the instruments and develop appreciation for the Himig Kawayan ensemble.

C. Presenting examples/instances of the new lesson

1. Tell the students that they will be watching a video of a Himig Kawayan group. Ask them to identify and write down the instruments that they will recognize in the video presentation.
2. Let the students watch a video on the performance of any of the following Himig Kawayan groups:
 - a. Pangkat Kawayan founded by Victor Toledo
 - b. Musikawayan founded by Gilbert Ramos
 - c. Banda Kawayan Pilipinas founded by Siegfredo Calabig

D. Discussing new concepts and practicing new skills # 1

1. Ask the students to study the list of instruments that they made and to classify them according to the general classification indicated on the board. Let the students explain the basis of the classification on the board. The students should be able to identify and classify the following:
 - a. Angklung – idiophone
 - b. Marimba – idiophone
 - c. Bumbong – aerophone
2. Ask students to explain how the instruments are played as well as the functions of the individual instruments in the context of the ensemble as presented in the videos. Discuss that there may be other bamboo instruments tuned diatonically or other percussive bamboo instruments included in the Himig Kawayan ensemble as seen in the videos.
3. Let the students watch tutorial videos on angklung, marimba and bumbong for more information.

E. Discussing new concepts and practicing new skills # 2

Option 1 - If there are available Himig Kawayan instruments

1. Identify the founders/manufacturers of the Himig Kawayan instruments and discuss their contribution to Philippine musical tradition.
2. Distribute the pieces of angklung to the students. Demonstrate the manner of playing the instruments and other playing techniques. Let the students play the C major scale in legato manner then in staccato manner.

3. Guide the students in learning and performing “Manang Biday” through any of the following techniques:
 - a. Reading the notes of the piece (if students are note readers)
 - b. Memory (through an assignment, ask the students to memorize the piece before this lesson on Himig Kawayan)
 - c. Teacher points at the players one note at a time
4. If the students can perform the angklung part on their own, play or ask anyone in the class to play the marimba part (and bumbong) as indicated in the piece. (The bumbong among the Himig Kawayan instruments is the most difficult to play. However, it can easily be learned through correct formation of the lips. Refer to the video on bumbong tutorial.)

Option 2 - If there are no available Himig Kawayan instruments

1. Identify the founders/manufacturers of the Himig Kawayan instruments and discuss their contribution to Philippine musical tradition.
2. Let the students watch more videos on:
 - a. Individual Himig Kawayan instruments
 - b. Himig Kawayan ensemble presentations

F. Developing mastery (leads to a formative assessment)

Option 1 - If there are available Himig Kawayan instruments

Have the students perform the first part or the whole piece of “Manang Biday” (depending on their skill or capacity). Let them practice playing the music, observing good phrasing and articulation with the end goal of achieving beautiful music (within the bounds of the time allotment for the class).

Option 2 - If there are no available Himig Kawayan instruments

Ask the students to identify the other instruments in the Himig Kawayan ensembles aside from the basic instruments discussed earlier. Let the students describe how the manner of playing and the material used in the instrument affect the quality of sound of the instruments.

G. Finding practical applications of concept and skills in daily living

Ask the students how playing together in an ensemble of angklung/ Himig Kawayan can be likened to the members of a family or to the different parts of our body in terms of function, contribution, or role.

H. Making generalization and abstraction about the lesson

Ask the students to: identify the basic instruments of the ensemble; describe the functions and the manner the instruments are played; identify the founders/manufacturers of the Himig Kawayan ensembles and their contribution to the Philippine musical tradition.

I. Additional activities for application remediation

Option 1 - If there are available Himig Kawayan instruments

1. Assign students to their respective Himig Kawayan instruments and ask them either to finish working on “Manang Biday,” part A and B (if the whole piece was not played during the class because of time constraints); or practice refining the whole piece with the correct interpretation.
2. Have the students perform the piece in the following meeting for a practical test (using rubrics on instrumental performance).
3. As a supplemental activity: Teach the students how to perform “Pambansang Awit” in another session.

Option 2 - If there are no available Himig Kawayan instruments

Assign the students to research on the history of any of the groups of Himig Kawayan listed below. For the next meeting, ask students to present their findings to the class.

1. Pangkat Kawayan founded by Victor Toledo
2. Musikawayan founded by Gilbert Ramos
3. Banda Kawayan Pilipinas founded by Siegfredo Calabig

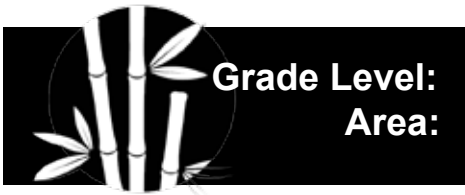
V. TEACHER'S REMARKS

VI. REFLECTION

- A. How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
- B. Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
- C. Which of my teaching strategies worked well? Why did this work?
- D. What difficulties can my principal or supervisor help me solve?
- E. What innovation or localized materials did I use/discover which I wish to share with other teachers?

KARATONG FESTIVAL OF DULAG, LEYTE

Prepared by: JOCELYN TIMBOL-GUADALUPE



Grade Level:
Area:

5
Music

I. OBJECTIVES

A. Content Standards

1. Recognize the musical symbols and demonstrate understanding of concepts pertaining to rhythm
2. Demonstrate understanding of the uses and meaning of musical terms in form
3. Demonstrate understanding of variations of sound density in music (lightness and heaviness) as applied to vocal and instrumental music

B. Performance Standards

1. Perform with a conductor a speech chorus in simple time signatures
 - a. With a choral ensemble
 - b. With an instrumental ensemble
2. Create and perform a song with appropriate musicality
3. Participate in a group performance to demonstrate vocal and instrumental sounds

C. Learning Competencies/Objectives

1. Identify aurally and visually different instruments in bamboo group/ensemble including local indigenous ensembles
2. Describe how music contributes to the performance of the musical production
3. Explore ways of producing sounds on a variety of sources similar to instruments being studied
4. Improvise simple rhythmic/melodic accompaniments to selected music from the Visayas
5. Evaluate music and music performances applying knowledge of musical elements and style
6. Create a variety of sounds emanating from the environment using available sound sources
7. Recognize rhythmic patterns using quarter note and eighth note in simple time signatures
8. Create different rhythmic patterns and ostinato using notes and rests in quadruple meter.
9. Recognize the design or structure of simple musical forms:
 - a. Unitary (one section)
 - b. Strophic (same tune with 2 or more sections and 2 or more verses)

II. CONTENT

- A. Rhythm - rhythmic patterns using triplets in meter in 4
- B. Form - structure of musical sound: Strophic
- C. Timbre - vocal and instrumental sounds
 - 1. Speech choir
 - 2. Karatong (bamboo slit drum) ensemble

III. LEARNING RESOURCES

A. References

- 1. Print Materials
 - a. BMI ethnography of Dulag's Karatong Festival
 - b. BMI documentation pictures of the instruments used in the karatong ensemble
- 2. Audio-Video Materials

BMI documentation videos of Dulag 75th Anniversary Celebrations of MacArthur Landing featuring:

 - a. Labiranan Festival of the 2019 Karatong Festival by San Jose National High School, Dulag, Leyte
<http://tiny.cc/BMI-Dulag-Karatong>
 - b. "Commitment to Peace" speech choir presentation by the Youth of Dulag
<http://tiny.cc/BMI-Dulag-Choir>
- 3. Music Scores

"Si Felimon"

B. Other Learning Materials

- 1. Classroom percussion instruments
- 2. Other found objects (canisters, plastic containers for drums, small bottles with beans as shakers)

IV. PROCEDURE

A. Reviewing the previous lesson or presenting the new lesson

- 1. Show a picture of a fisherman.
- 2. Have the students watch and listen to a rendition of "Si Felimon" excerpt by the San Jose National High School karatong performers.
- 3. Tell them of the background of the song and the story behind the presentation (story of a fishing village that was able to fight off spirits that threatened their fisherfolk through the power of the patron saint and the music of the karatong).

B. Establishing the purpose of the lesson

Tell the students that after a review of rhythm in the First Quarter, they will now integrate their rhythmic learning with form and timbre by composing their own rhythmic patterns accompanying the song.

C. Presenting examples/instances of the new lesson

1. Listen and Imitate: Have the students listen to the karatong rhythm and imitate it.
2. Present the notation after successful clapping of the pattern.

D. Discussing new concepts and practicing new skills #1

1. Let the students sing the song following the notation and lyrics of the song.
2. Divide the class into three with one group tapping the beat on their laps, another group clapping the rhythm/words of the song, and another group singing the song.
3. Using listening for rhythm and tonal patterns, have the students describe the structure of the phrases/lines of the song (verses with 4 lines following aabc).
4. Introduce the term STROPHIC as the term that describes the form of the song where the tone and rhythm are repeated in each verse of the song.

E. Discussing new concepts and practicing new skills #2

1. Let the students watch the video and look at the instruments used once more.
2. Show pictures of the instruments as appropriate.
3. Find equivalent instruments in the classroom for low percussion sounds (drum), middle percussion (canisters), and high percussion (shakers).
4. Guide the students in composing 4 lines of 8-beat ostinato in 4/4 using classroom percussion instruments applying the phrase pattern of the song: aabc. Rhythmic notation may be used as a review.

F. Developing mastery (leads for formative assessment)

Let the students practice their composed ostinato to accompany the song.

G. Finding practical applications of concepts and skills in daily living

Have the students watch a speech choir use various rhythms in their speech choir rendition.

Prompt: How can you use rhythm and form in your daily speech to communicate effectively?

H. Making generalization and abstraction about the lesson

1. Using shapes, represent the form of each line of the song.
Example circle for a, triangle for b, square for c.
2. Conclude by saying that form in music is similar to shapes. Using lines, one can draw different shapes. Similarly, using tones and rhythms played on instruments, one can make various forms of sound.

I. Additional activities for application or remediation

1. Beat competence: Students with limited beat competence can be asked to move to the beat using bigger movements such as stepping to the beat while singing the song.
2. Tonal competence: Students with limited singing ability can tone match simple one to two tones with a classmate/teacher.

V. TEACHER'S REMARKS

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VI. REFLECTION

- A. How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
- B. Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
- C. Which of my teaching strategies worked well? Why did this work?
- D. What difficulties can my principal or supervisor help me solve?
- E. What innovation or localized materials did I use/discover that I wish to share with other teachers?

MUSIKONG BUMBONG OF ANGONO, RIZAL

Prepared by: DECEMBER RAGRARIO-VALENCIANO



Grade Level:
Area:

5
Music

I. OBJECTIVES

A. Content Standards

Recognize the musical symbols on the element of Tempo and demonstrates understanding of these concepts pertaining to speed in music

B. Performance Standards

Apply appropriately, various tempo to vocal and instrumental performances

C. Learning Competencies/Objectives

1. Identify the various tempo used in music heard
2. Use appropriate musical terminology to indicate variations in tempo
 - a. Largo
 - b. Presto
 - c. Allegro
 - d. Moderato
 - e. Andante
 - f. Vivace
 - g. Ritardando
 - h. Accelerando
3. Employ appropriate tempo in the vocal and instrumental performance of a given music

II. CONTENT

Musikong Bumbong of Angono

III. LEARNING RESOURCES

A. References

1. Print Materials

- a. BMI ethnography of Angono's Musikong Bumbong
- b. Pictures of Angono Musikong Bumbong from BMI documentation
- c. Pictures of Gigantes of Angono Fiesta
- d. Textbook pages
- e. GIT/GUHIT ANGONO
p. 288-289 (c) 2018, Editor: Richard R. Gappi
ISBN 978-971-95770-5-8, Volume VIII. Issue 30.
Nov. 18-25, 2018. ISSN 2244-3851.
DTI Registration No. 01987812.
Exponent of Community Journalism.
- f. Angono Rizal News Online
November 15, 2019.
Culture/Heritage: On Angono Fiesta's Parehadoras, Endramada and Bands of Musicians

2. Audio-Video Materials

- a. BMI interview with Prof. Jose Rommel "Jopat" Gragera, conductor of Angono Wind Ensemble, and Angono Bamboo Instrument Maker
<http://tiny.cc/BMI-Gragera-Interview>
- b. Music videos from YouTube:
 - Video of The Swan: Musikong Kawayan Bamboo Ensemble
 - Video of Pilipinas Got Talent Season 5 Auditions:
Alicia Bohol Musika Kawayan - Bamboo Musicians
 - Video of Angono Wind Ensemble

B. Other Learning Materials

1. To make rainstick: empty thick cardboard roll, beans, small nails, small hammer, packaging tape, crayons/ribbons for decoration
2. Other available bamboo instruments i.e., bamboo sticks, triangles, tambourines

IV. PROCEDURE

A. Reviewing the previous lesson or presenting the new lesson

1. Show a picture of Higantes.
2. Ask where they are popular and what they are associated with. (Higantes are popular during the fiesta in Angono, Rizal and are said to represent the artistry of Angono).

B. Establishing the purpose of the lesson

This is what we will learn about today:

1. About some information on the music tradition of Angono, Rizal
2. How the speed of a music or song is important in the way the music is performed
3. Singing and interpreting a song composed by a National Artist born in Angono

C. Presenting examples/instances of the new lesson

1. Show audio-videos

- a. Music #1:
The Swan: Musikong Kawayan Bamboo Ensemble
- b. Music #2:
Pilipinas Got Talent Season 5 Auditions:
Alicia Bohol Musika Kawayan – Bamboo Musicians

2. Discussion:

What instruments are playing?

What could these instruments be made of? (bamboo)

How do you feel about Music #1? (peaceful – swan swimming)

(Teacher gives title of music and event)

With your answer, could this be why the music is played slow? How do you feel about Music #2? (dance – tinikling)

(Teacher gives title of music and event)

With your answer, could this be why the music is played fast?

D. Discussing new concepts and practicing new skills #1

1. Define and discuss tempo in Music
2. Present and discuss music terminologies for tempo
3. Relate terminologies for tempo with music listened to a while ago.
4. Show video of “Ugoy ng Duyan” by Lucio San Pedro and sang by Lea Salonga
5. Teach the song via Teaching a Song Method:
 - a. Listen to whole song
 - b. Brief discussion on context of song:
 - Music composed by National Artist Lucio San Pedro from Angono, Rizal
 - Inspiration for composition is composer’s mother who used to hum some part of the melody when putting her children including Lucio, to sleep.
 - c. Rhythmic reading/syllabication by phrase
 - d. Learning the melody by phrase
 - e. Sing the whole song
6. In groups, students will choose from among the presented Music Terminologies for Tempo that are most appropriate to apply in the performance of the song.

E. Discussing new concepts and practicing new skills #2

1. Show photos of the Angono bamboo musical instruments
2. Show the video:
Bulacan Bamboo Instruments with Angono bamboo instruments:
Video of Angono Wind Ensemble
3. Discuss

On Video 1:

Upon hearing music which you identified as being performed by bamboo instruments. The instruments seen in the photos and videos such as bamboo baritone, French horn and trombone are called Musikong Bumbong instruments which are famous in Bulacan and are in fact still existing and performed. Since we will be talking about music from Angono, there are however no more Musikong Bumbong instruments that still exist save for very few that are displayed in the home of the maker’s family. The difference between the Musikong Bumbong of Bulacan and the Musikong Bumbong of Angono is that those from Angono do not have the woven rattan bells but instead have buzzer end pipes.

The tradition of the famous bands of Angono is said to have started with a group of musicians who played musical instruments made of bamboo because they could not afford to buy those expensive brass instruments. These brass instruments were called Musikong Bumbong or Band instruments made of bamboo. With their strong musicianship, they created their own instruments out of the bamboo that was after all many in Angono. One of its makers was Lieutenant Lauriano Gragera His instruments no longer exist, but there are a few displayed on the wall of a relative’s house and are covered with cling-wrap for protection and posterity. These instruments are now said to be about 60 years old.

On Video 2:

Bands are very important in the Philippines because they are specially played during fiestas. In Angono, the fiesta is called Feast of St. Clement, the Patron of Fishermen, celebrated on November 23; This fiesta is about thanking God for the blessings and bounty of the sea; Includes the famous Higantes or giants composed of a Father, Mother, and Son which represents Angono as the Arts Capital of the Philippines, because indeed the people of the place are very good in the Arts.

Bands play a major role on this festive day where each school or barangay has a band that joins to help fill and keep the joyful mood and high spirit of occasion and for every fiesta occasion, at least 10 bands go marching and playing their music around the town.

F. Developing mastery (leads for formative assessment)

1. Group Performance Assessment on application of Tempo, students will perform the song “Sa Ugoy ng Duyan” employing appropriate Tempo markings with accuracy and creativity. Available instruments i.e., bamboo sticks, rainsticks, triangles, tambourines are allowed to add texture to the performance of the song.
2. Assessment with teacher-made rubrics

G. Finding practical applications of concepts and skills in daily living

Prompt: To manifest value for the environment, students can make homemade instruments, i.e., rainstick made out of an empty cardboard roll (instead of bamboo), beans, packaging tape.

With siblings at home, students can create a soundscape with the homemade rainstick, producing slow and peaceful as well as fast and furious imaginations.

H. Making generalization and abstraction about the lesson

Music is one influential form of showing proper and creative use of one's environment just as Lieutenant Lauriano Gragera did.

I. Additional activities for application or remediation

For reinforcement on Lesson, Since Angono is the Art Capital of the Philippines, and the Higantes is the symbol that represents the artistry of Angono, students will create an artwork reflecting on their understanding of the song “Sa Ugoy ng Duyan” employing the theme on family of the Higantes.

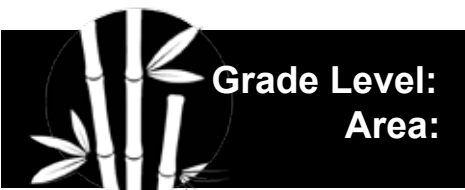
V. TEACHER'S REMARKS

VI. REFLECTION

- A. How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
- B. Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
- C. Which of my teaching strategies worked well? Why did this work?
- D. What difficulties can my principal or supervisor help me solve?
- E. What innovation or localized materials did I use/discover that I wish to share with other teachers?

MUSIKONG BUMBONG OF BULACAN

Prepared by: DECEMBER RAGRARIO-VALENCIANO



I. OBJECTIVES

A. Content Standards

Recognize the musical symbols and demonstrates understanding of concepts pertaining to volume in music

B. Performance Standards

1. Visually and aurally apply appropriate dynamics to a given musical selection
2. Sing expressively, with appropriate dynamics, phrasing, and interpretation

C. Learning Competencies/Objectives

1. Identify the different dynamic levels used in a song heard
2. Apply appropriate musical terminology to indicate variations in dynamics in a performance of a song, specifically: piano (p), mezzo piano (mp), forte(f), mezzo forte (mf), crescendo (<), decrescendo (>)
3. Write instances when the Dynamics of family life is experienced and valued.

II. CONTENT

Musikong Bumbong of Bulacan

III. LEARNING RESOURCES

A. References

1. Print Materials
 - a. BMI ethnography of Bulacan's Musikong Bumbong
 - b. Pictures of Bulacan Musikong Bumbong from BMI documentation such as baritone and French horn
 - c. Pictures of a lone duck/swan swimming in the lake; children marching
 - d. Textbook pages

2. Audio-Video Materials

- a. BMI interview with Rodrigo Anastacio, Bulacan Bamboo Instrument Maker
<http://tiny.cc/BMI-Bulacan-Anastacio>
- b. BMI documentation of Marcelo H. Del Pilar students playing bumbong at Hiyas ng Bulacan 2020, Malolos, Bulacan
<http://tiny.cc/BMI-Bulacan-MHDP>
- c. Music videos from YouTube:
 - Video of Sta. Clara Pinung-pino - Fertility Ritual/Music & Dance of Bulacan - NAMCYA 2012 – CCP
 - Video of Katrina Marzella on the Baritone: The Swan (Il Cigno)
 - Video of Grade School French Horn Solo Ensemble

3. Music Scores

“Sta. Clara Pinung-Pino,” folk song of Bulacan

B. Other Learning Materials

Available Bamboo Instruments i.e., tongatong (bamboo stamping tube), bungkaka (bamboo buzzers), triangles, and tambourines

IV. PROCEDURE

A. Reviewing the previous lesson or presenting the new lesson

1. Show video of “Sta. Clara Pinung-pino” – Fertility Ritual/Music & Dance of Bulacan - NAMCYA 2012 – CCP
2. Ask/Mention:
 - a. Have you heard of this festival in Bulacan?
 - b. This is the Obando Dance, a festival in the province of Bulacan where childless couples request through prayer and music to give them a child and the opportunity to have a complete family. This festival is a manifestation of the Filipinos’ value for family.
 - c. Get experiences from students re value for family life.
 - d. Let’s check on this festival: What do you observe in the way the song is sung? (Expected answer: fervently)
 - e. What do you observe in the way the Refrain “*Abaruray, abarinding, ang pangako’y tutuparin*” is performed/danced? (Expected answer: festively, enjoyably)

B. Establishing the purpose of the lesson

This is what we will learn today:

1. Singing and interpreting the song
2. We will also see musical instruments from Bulacan that are made of bamboo
3. Then we will write about family life.

C. Presenting examples/instances of the new lesson

1. Matching pictures (a, b) with musical examples (c, d): picture (a) lone duck or swan swimming in a lake, and picture (b) children marching
2. Listening to musical examples c and d: Katrina Marzella on the Baritone: The Swan (Il Cigno) and Grade School French Horn Solo Ensemble:
3. Ask students why picture (a) with music (c) and picture (b) with music (d) are the correct answers.

D. Discussing new concepts and practicing new skills #1

1. Mention Dynamics as the reason behind the answers to the previous activity
2. Define and discuss Dynamics in Music
3. Present and discuss music terminologies for Dynamics
4. Present pictures of Bulacan bamboo musical instruments:
5. Present the bamboo-made baritone and French horn that they just heard that used the Western baritone and French horn:
6. Briefly discuss the tradition, value of producing, system/technique of playing the instruments of the musikong bumbong of Bulacan
7. Ask how loud or soft can such instruments produce? (usually loud-forte). Continue to discuss that such instruments can produce any degree of loudness or softness depending on how expressive the performer would be just like the musical examples they just heard.

E. Discussing new concepts and practicing new skills #2

1. Present song "Sta. Clara Pinung-Pino" 1st verse with refrain.
2. Learn song via *Teaching a Song Method*.
 - a. Listen to whole folk song
 - b. Brief discussion on some characteristics of folk songs:
 - Talks on social issues
 - Learned through oral tradition
 - Intent of composer is obviously not for profit that is why composer is usually not known
 - c. Rhythmic reading/syllabication by phrase
 - d. Learning the melody by phrase
 - e. Sing the whole song
 - f. Practice song in groups employing expressive elements

3. Recap from previous observations on video: how the 1st verse of song is sung (fervently as it is a prayer), and how the refrain is interpreted (festive as it is usually part of the dance/festivities).
4. Allow students to choose from among the presented music terminologies for Dynamics that are most appropriate to apply when singing the 1st verse and the refrain.

F. Developing mastery (leads for formative assessment)

1. Students perform the 1st verse of “Sta. Clara Pinung-Pino” using piano/ soft dynamics (fervent interpretation) and the Refrain using crescendo dynamics (swelling in more and more festive mood).
2. For Group Performance Assessment on application of Dynamics, the students employ as much creativity and variations in applying the concept of Dynamics. For the refrain part, adding available instruments like bamboo instruments i.e., tongatong (bamboo stamping tube), bungkaka (bamboo buzzers), triangles, tambourines are allowed.
3. Assessment with Teacher-made Rubrics

G. Finding practical applications of concepts and skills in daily living

Prompt: At home, one experiences the dynamics of family life. Write some instances/experiences when these happen i.e., praying together – piano; healthy misunderstandings with siblings - forte; eating together – crescendo decrescendo (talking while eating becomes intense then decreases depending on topics); etc.

H. Making generalization and abstraction about the lesson

On fun days when your family gets the opportunity to go to Jollibee, do you order/like Chicken Joy because it is delicious? Performing music is like eating chicken joy. We perform a song expressively, so we feel good just as eating the Chicken Joy slowly to taste its deliciousness makes us feel so good as well. If we sing or perform a song without expression, just as if we eat the Chicken Joy hastily, we do not enjoy it. That is why Dynamics in music is important because it helps us in the expressive element/aspect of the music.

I. Additional activities for application or remediation

1. For reinforcement on Lesson, show BMI Interview with Mr. Rodrigo Anastacio at Hiyas ng Bulacan, Malolos, Bulacan
2. Question: What are some values we can get from Mr. Anastacio’s advocacy of continuing on with the tradition of the Musikong Bumbong in Bulacan? (Some expected answers: keeping one’s tradition and culture, care for environment)

V. TEACHER'S REMARKS

VI. REFLECTION

- A. How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
- B. Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
- C. Which of my teaching strategies worked well? Why did this work?
- D. What difficulties can my principal or supervisor help me solve?
- E. What innovation or localized materials did I use/discover that I wish to share with other teachers?

TULTUGAN MUSIC OF ILOILO

Prepared by: HERMIE F. CARTAGENA



Grade Level:
Area:

5
Music

I. OBJECTIVES

A. Content Standards

1. Recognize the various rhythmic patterns and show an understanding of these concepts pertaining to rhythm
2. Recognize and differentiates the various melodic contours and show an understanding of the musical elements pertaining to pitch
3. Recognize the different tempo and show understanding of these concepts related to speed

B. Performance Standards

1. Perform various rhythmic patterns individually and with a group
2. Perform various melodic contours individually and with a group
3. Perform various melodic and rhythmic patterns in an ensemble implementing various tempos

C. Learning Competencies/Objectives

1. Aurally and visually identifies and differentiates the instruments of the tultugan
2. Improvise on various rhythmic patterns reflecting the musical characteristics of tultugan music
3. Improvise on various melodic contours reflecting the characteristic of tultugan music
4. Creates melodic and rhythmic patterns in different tempo (fast and slow)

II. CONTENT

- A. Rhythm - rhythmic patterns using quarter, eighth, sixteenth, and dotted notes in common and triple time signature
- B. Melody - melodic contours and directions (high and low)
- C. Tempo - Instrumental ensemble using various tempo or speed

III. LEARNING RESOURCES

A. References

1. Print Materials

- a. BMI ethnography of Maasin's Tultugan Festival
- b. Pictures of tultugan instruments and performance

2. Audio-Video Materials

YouTube videos of performances in Tultugan Festival

3. Music Scores "Tatlo ka Pato"

B. Other Learning Materials

Other Learning Materials

Improvised percussion instruments (empty bottles, cup, plastic containers, empty can, PVC pipes)

IV. PROCEDURE

A. Reviewing the previous lesson or presenting the new lesson

1. Show a picture of a bamboo plant.
2. Elicit answers on the various functions and uses of bamboo.
3. Ask the students to find the Visayas and locate Iloilo province on the Philippine map.
4. Show pictures of Maasin's Tultugan Festival and ask the students about their first impression.

B. Establishing the purpose of the lesson

Inform the students that tultugan music contains various rhythmic and melodic elements in music. Tell the students that they will be demonstrating these concepts through their personalized performance.

C. Presenting examples/instances of the new lesson

Listen and Imitate:

1. Teacher claps a rhythmic pattern and students imitate the teacher.
2. Have the students listen to the tultugan rhythm and imitate it.
3. The teacher presents stick notations on the board and clap the patterns, and the students imitate the teacher.
4. Let the student create a rhythmic pattern by clapping and writing its stick notation.

D. Discussing new concepts and practicing new skills #1

1. Let the students learn the song “Tatlo ka Pato” aurally/orally.
2. Let the students sing the melody of the song while clapping its rhythm.
3. Teacher presents the notation of the song.
4. Teacher points on the notation while students clap the rhythm of the song as they sing the melody.
5. Sing the song in a different tempo (slow and fast).
6. Students improvise on movements as they sing the song “Tatlo ka Pato.”

E. Discussing new concepts and practicing new skills #2

1. Show a picture of different instruments related to tultugan.
2. Let the students identify the role of each instrument in the ensemble.
3. Let the students find relevant or equivalent instruments in the classroom (melodic and percussion).
4. Instrumental performance: Let the students play the song “Tatlo ka Pato” using their improvised instruments (melodic and percussion).

F. Developing mastery (leads for formative assessment)

1. Let the students compose their own rhythmic and melodic patterns individually and with a group.
2. Divide the class into three groups: group 1 will sing the melody, group 2 will improvise on movements, and group 3 will improvise on the instrumental accompaniment.
3. Let the students experience the different tempo (slow and fast).

G. Finding practical applications of concepts and skills in daily living

1. Read a poem written in the students’ language and let them find the relevance of rhythm to speech.
2. Listen to slow and fast music and let the students realize how the music influences our emotions or feelings.

H. Making generalization and abstraction about the lesson

Rhythm, melody, and tempo are musical elements that create musical ideas and musical expressions.

I. Additional activities for application or remediation

For reinforcement: Let the student explore different rhythmic and melodic patterns using different instruments. Encourage students to experiment with various sounds by making a personalized bamboo instrument.

V. TEACHER'S REMARKS

VI. REFLECTION

- A. How many learners earned 80% in the evaluation? How many learners require additional activities for remediation?
- B. Did the remedial lesson work? How many learners have caught up with the lesson? How many learners continue to require remediation?
- C. Which of my teaching strategies worked well? Why did this work?
- D. What difficulties can my principal or supervisor help me solve?
- E. What innovation or localized materials did I use/discover that I wish to share with other teachers?

Appendix A – Ethnographies

Excerpt from the BMI Ethnography of Las Piñas Bamboo Organ

“BMI Project: The Las Piñas Bamboo Organ”

By Juliet R. Bien

INTRODUCTION AND HISTORICAL BACKGROUND

To date, the Las Piñas Bamboo Organ is the oldest and the largest in the Philippines, and possibly the world (Tagle, interview, 9 February 2020). This musical instrument is the one and only functional organ made of bamboo and is actively used in liturgical services. What sets this apart from the other existing church organs is not only the material from which it is made but also its endurance because it has been around for more than two hundred years. The bamboo organ of Las Piñas is a cultural treasure, the only one of its kind in the Philippines.

At one point in the history of music, the organ was considered as the “king of instruments” because of its many musical characteristics together with its often grandiose presentation in churches. Its builder was not only a good musician, but also a universal genius with extraordinary talents as wood sculptor, metal-smith, painter, mathematician, and physicist. In the case of the Bamboo Organ, its builder was more than all that. Diego Cerra was also a natural scientist whose feat in curing the bamboo pipes was never successfully duplicated in decades and centuries to come. (Lauterwald, Helen Samson. The Bamboo Organ of Las Piñas, Manila: The Bamboo Organ Foundation, Inc., 102).

The Bamboo Organ was made by Father Diego Cera within the period of his assignment as parish priest of Las Piñas from 1795 to 1832. He began collecting the bamboo in 1816 and buried them in the seashore. The bamboos were unearthed in 1817 and by 1821, the organ was almost finished (Lauterwald, p.97). The long process of instrument-making required the cooperation of and assistance of local craftsmen who were already working with bamboo used for furniture and other household items (Tagle, interview, 2020). The exact date of installation is not known but it is believed to have taken place in 1824 when it was presumably completed.

The construction of the instrument was complicated because funding had to be sourced from Father Cera's superiors and indispensable parts such as sheepskin, which were not locally available, had to be shipped. It was revolutionary and experimental at the same time because it would be made almost completely of bamboo. Historical records reveal that locals helped Father Cera in making the instrument.

In 1816, the bamboo was harvested and buried in the seashore. In 1817, Father Cera unearthed the bamboo and started constructing the instrument. Other wood types like narra, molave and kamagong were used for the mechanism as well. Simultaneously, the bellows were constructed, and sheepskin was shipped into the county. Last to be installed were the metal pipes which are believed to have come from Spain.

The entire set-up of the Bamboo Organ is like a house with several rooms installed in the wall of the Church. To access the instrument, one has to ascend the choir loft and descend a narrow flight of stairs. The organist plays from a small narrow

porch as if suspended but stable with the congregation approximately fifteen to twenty feet below. The chambers are on the opposite wall unseen to the audience but can be accessed by ascending a narrow wooden 5-foot ladder which can accommodate a single person. Upon reaching the topmost step, one enters a wooden room approximately 15 square meters which houses the mechanism, a hollow space surrounded by the old non-functional pipes and the intricate network of pipes from one end to the other. The keyboard levers could be seen on the first floor of the chamber together with the *pajarillo* mechanism, which has to be partially filled with water to provide the bird-like sound. Going up a few steps of a wooden ladder, one can see the other side of the bamboo pipes and the reed pipes.

The Bamboo Organ is now permanently installed on the upper left wall of the parish church of Saint Joseph, adjacent to the choir loft. As a celebration of heritage and sharing the instrument with the world, the Bamboo Organ Festival, a gathering of international and local musicians is held annually during the month of February. Repertoire for the organ is dedicated in these concerts which began in 1976 and continues to the present.

CEALWYN TAGLE: CONSERVING AND MAINTAINING THE BAMBOO ORGAN

Cealwyn Tagle's involvement with the bamboo organ began when he was a member of the Las Piñas Boys Choir when he was nine years old. Leo Renier, the choirmaster at the time, fostered his interest in the organ and encouraged him to take care of the instrument when Cealwyn was sixteen years old. Cealwyn had just graduated from high school when Mr. Renier referred him to Fr. Johann Trummer, a priest, organist, and a supporter of the Bamboo Organ Festival. Looking at the exquisite instrument as "orphaned" and exposed to neglect, Cealwyn accepted the job totally clueless of what the future had in store for him. Abandoning his desire to continue his college education, he bade his mother goodbye and left for Austria in 1988 to be an apprentice in organ making for four years. Fr. Trummer looked after Cealwyn, Armando Salarza (current Las Piñas Bamboo Organ resident organist), and other scholars from Las Piñas during their stay in Austria, and it was him who made their studies possible.

Cealwyn's interest in his new "job" fueled his desire to be in this field because of the country's lack of experts in organ conservators. His exposure to the European culture's utilization of organs in churches could have also increased his interest to promote the presence of organs in the numerous churches in the Philippines. After his four-year stint, Cealwyn stayed for two more years in Germany and worked for the company of Helmut Allgauer Orgelbau, where he was exposed to various fields in organ building. Here, his initial exposure as an apprentice was challenged with the intensive all-around work in the company. Cealwyn considers this period fortunate because, Orgelbau's company being small, he was always assigned to the difficult task of voicing the organs the company made. While improving his skills in woodworking, installations and repair, voicing enabled Cealwyn to hone his aural skills and craftsmanship. All these experiences have helped him in maintaining the Bamboo Organ and have also equipped him with the necessary skills needed in setting up an organ manufacturing company for himself upon his return to the Philippines.

Cealwyn came back to Las Piñas and was immediately assigned as the administrator of the Bamboo Organ. As administrator – a position he still holds to this day – he is tasked to conserve, repair, and maintain the organ. From the nine-year-old chorister during the 1970's, Cealwyn is now in his 40's, a family man – married with three children – and the custodian of a 200-year-old natural cultural treasure, a post he never imagined occupying in his wildest dreams.

Cealwyn believes that Father Cera was sent to the Philippines primarily as an instrument-maker and secondarily as a parish priest. Father Cera most probably grew up as an organ builder before he was sent to the seminary and could have been sent to the Philippines to repair the organ at the Manila Cathedral. Aside from this, Father Cera also built the Baclayon Church in Bohol, whose flat façade feature is his signature style. It is also worth noting that Cealwyn is a direct descendant of Calixto Lara, one of the local citizens of Las Piñas who worked with Father Cera in maintaining the instrument together with his two brothers Pedro and Juan. The involvement of Calixto Lara's family with the Bamboo Organ now comes full circle as his great-great-grandson Cealwyn is now also in-charge of the Bamboo Organ after a 200-year interval.

Excerpt from the BMI Ethnography of Cebu's Bamboo Ukulele

"The Bamboo Ukuleles of Cebu: A Trailblaze"

By Yves P. Cuenza

THE UKULELE BAMBOO INSTRUMENTS

There are different factors as to why a lot of people (both musicians and non-musicians) tend to get attached to playing the ukulele. Mr. Brian Brusas of the Huni group asserts that the advantages of the ukulele instrument include its being small and handy. Moreover, the laid-back perception that it gives does not intimidate or give pressure to the musical ability of the person playing it.

The *Bamboo Musical Instruments (BMI) Description Table* below illustrates the differences of each ukulele:

Instrument		Method of Sound Production	Size (scale length)	Number of frets	Tuning	Role in the Ensemble
ukulele	sopranino	Ukulele is a lute instrument that is plucked and strummed	28 cm	10	d1 g1 b1 e2	It can be used as a solo instrument, for accompaniment, and for ensemble music making.
	soprano		35 cm.	12-15	g1 c1 e1 a1	
	concert		38.5 cm	15-20	g1 c1 e1 a1	
	tenor		43 cm	15-20	g1 c1 e1 a1	
	baritone		51.5 cm	18-19	d g b e 1	

The Huni group produces five types of ukuleles. The *soprano*, *concert*, and *tenor* are the most commonly used ukuleles which are tuned on the same key but gives give different over-all timbres. The difference is usually seen on the body size which is proportionate to the sound it gives off – the bigger the size, the deeper and fuller the ukulele would sound. The *baritone* is the largest among the ukuleles and it has the same tuning with the upper four strings of the guitar. The smallest is the sopranino and has the same tuning with the baritone but is set one octave higher. The price of every bamboo ukulele ranges from P5,000 to as much as P10,000, depending on personalized requests from the buyer.

The shop has been producing handmade bamboo ukuleles. The most preferred species as raw material is locally known as *bontong*. If a customer wishes to get a bamboo ukulele, they must know the advantages and disadvantages of having one. Mr. Brusas and his team are quite upfront and honest in presenting those terms. Probably the only disadvantage about the sound is that the bamboo material does not necessarily produce the brightest timbre compared to the more traditional materials like mahogany, mango and *langka* woods. On the other hand, one of the strongest feedback that the bamboo material has been getting comes from foreigners, who say that the bamboo ukulele is more adaptable to climate and seasonal changes wherever the instrument is taken.

Excerpt from the BMI Ethnography of Quezon City's Pangkat Kawayan

Musiko Bumbong of Pangkat Kawayan: The Glory of the Singing Bamboos of the Philippines

By Janine Josephine Arianne A. Liao

A LIFETIME GUARANTEE: ELEVATING THE HUMBLE BAMBOO TUBE

Under the creative leadership of Victor Toledo, the old Pangkat Kawayan of Quezon City produced music using the bamboo instruments that were created, designed, and refined by the master himself. The current Pangkat Kawayan uses the same instruments that were made at least thirty years ago but have aged beautifully with the group. With the dedication and attention to detail evident in their quality, these instruments attest to the standard of a lifetime's guarantee that can only come from meticulous research and production.

One important aspect that I have noticed is that a major driving force for the creation of these instruments was for the bamboo instrument to be able to play a western, diatonic scale. In an article, Vic Toledo writes that the bumbong, or tuned bamboo tube, needs to play a complete scale, in tune, in order to make a beautiful sound. According to him, most bamboo instruments in the Philippines are "not tuned," that is to say, pitches are indefinite in comparison to western tuning. This undertaking to tune the bamboo seems to stem from a very cosmopolitan outlook, indicating a desire to participate in a global music culture. It also indicates a strong local thrust in developing distinctively Filipino instruments through utilizing and perfecting bamboo instrument making and design.

Among the instruments that the Toledo makers crafted, the most unique and significant is the *bumbong*. These are thick in diameter and appear to be quite heavy. With a hole at the side, it appears as a thick transverse flute. However, unlike the thin flutes with multiple holes, the bumbong produces a single pitch per tube. They have a complete set of two and a half octaves. As Vic writes, it is a unique contribution of the Philippines to the world.

Another instrument that looks unique is the *tipanklung* or bamboo piano. It is similar to the *angklung*, or bamboo rattles from Indonesia. However, instead of these played separately by different individuals, the rattles are hanged on a frame and arranged like a piano, including the raised black keys. It looks like a curiosity, but once it is hanged like so, it is possible to be played by one person. When they performed for us, this instrument carried the main melody. This instrument is one of the later creations of Vic Toledo.

They also have an instrument that they call the Philippine *angklung*, the *alugtog*, which is played similarly by groups in an interlocking style. However, the appearance of the instrument features a different design, the most notable of which is the three-line staff with a black sixteenth note in the center. This appearance follows the western model and is a mark that signals its easy adaptability to western-style music.

Another form of bamboo rattles are the chord rattles. What is unique about this instrument is that each rattle is already made up of three pitches, forming a chord, which can be played by one person as accompaniment. The members who demonstrated these performed with choreography, with simple turns and forward-backward sways.

They also have a marimba, whose bars are made of narra and resonators made of bamboo instead of metal. The marimbas in the group play melodies, harmonies, and obligato fills in the music arrangements. I noticed that the senior members of the Pangkat Kawayan were the ones who played the marimba.

An interesting instrument was lying at the corner of the *Pangkat* center. According to Beth, it was not used anymore. It was the *tunog hagdan* bamboo instrument. And true enough to its name, it looks like a ladder. The “steps” are made of narra bars, and can be classified as a xylophone instrument. Beth demonstrated the instrument for us by performing the song Pipit which was arranged by Vic Toledo. This instrument is unique to *Pangkat Kawayan* and is said to be one of Vic’s many musical instrument inventions.

Lying at the sides of the Pangkat center were the gabbang. The *gabbang* is a xylophone found among indigenous people in southern Philippines. However, according to Chino, these were not tuned and were therefore not used at all.

Finally, a curiosity of a piece was the *sungkit*. There was only one available, and as they said, it was for aesthetic purposes, and did not function for sound at all. It was a long piece of bamboo, similar to a *sungkit* or a device used to get fruit from trees.

The Toledo *bumbong* instruments have been around for a long time and are cleaned with alcohol as they have been shared all these generations. But these are indeed of good quality – well-aged, solid-sounding, and sturdy. For minor repairs, they just apply white glue, and for finetuning, a small piece of wood is attached to the inside, and I believe this small piece can then be manipulated to alter the tuning of the bumbong.

Currently, the sets that they use are stored in the Pangkat Kawayan center, but they can be easily packed for travel. Even the *tipanklung* can be disassembled for easy shipping. Aside from the ones used in performance, there are still some instruments around. These spares are stored in the Pangkat Kawayan center, but some are still at the Toledos’ house. There also may still be broken sets of *angklung*, still untuned. Sadly, those without storage space have been thrown away.

Meanwhile, a different form of invaluable wealth can be found in the Pangkat Kawayan Center: posted on the walls are articles on bamboo written by Victor Toledo himself. One of these articles is a collection of excerpts from a Japanese book on bamboo that he translated. Another is a treatise on the history of the *bumbong* (bamboo tube) as a cultural item and a musical instrument in the Philippines that he co-researched with a Vicente Toledo, professor of Spanish language and music in University of Manuel L. Quezon. Most noteworthy is this article that provides direct insight into the manufacturing process that Vic Toledo had mastered. An outcome of his personal experiences, research, observations, and consultations with builders and carpenters, this article discusses in detail the process of choosing and treating of bamboo, as well as cutting it and putting the pieces together to form instruments.

First, Vic shares that it is best to cut bamboo during summer. In the rainy season, the bamboos have new shoots, and these young sprouts have a “sweet” essence that attracts *bukbok* (rice weevils) which damage the bamboo. If cutting bamboo during the rainy season is unavoidable, it is best to soak it in vinegar in order to counter its

“sweetness” and prevent it from being attacked by *bukbok*. Once the bamboo is dried, wood treatments like Solignum can be used. Wood varnish – regular wood varnish, according to Vic’s son Chino – was also applied to the bumbong found in the Pangkat Kawayan Center, but Vic does not mention this in his articles. Also discussed is the use of sandpaper as a seal of sorts and how it provides better resonance, possibly due to the rough surface. This may be a unique practice to the Toledo makers.

Next, Vic warns against using old or mature bamboo. Older bamboo tubes, which can be identified by their telltale reddish tone, easily break and do not have the ideal sound. He says that the ideal age for bamboo for instrument making would be around five years, when bamboo possesses sturdy and strong fibers and a clear, bright sound. Clearly, he also cautions against using cracked, bent, or crooked bamboo tubes, as these cannot be tuned properly. He further shares that tropical bamboos have harder fibers, thus producing a bright and clear sound as opposed to bamboos that grow in more temperate regions, which in turn have softer fibers, therefore producing a softer and unclear tone.

Vic shares an interesting observation about how to tell if the bamboo is cut in the morning or at noon just by examining the “*buko*” or “eye” of the bamboo. He says that the *buko* is “closed” in the morning, but gradually opens up at noon time. He continues that that it is a trait of tropical bamboos but does not go any further to tell what this information may be used for. However, he enumerates four factors that affect the tuning of the *bumbong*: 1) the thickness or thinness of the bamboo; 2) the hardness and age of the bamboo; 3) the length, diameter, and the actual hole of the bamboo; and 4) the location or position of the hole for blowing into the tube. He emphasizes the importance of where the blow hole is bored into the bamboo tube as it affects the exactness of the pitch.

Chino recalls amusing stories of how his father, almost mesmerized by the beauty of the bamboos, would point at bamboos along the highway and remark if these were suitable for instrument making or not. This is truly a mark of the master craftsman who knows his materials well. In his articles, Vic mentions *bamboo kalabaw* (more commonly known and also referred to as *kawayang tinik* in the Pangkat Kawayan Facebook page). He refers to it with the scientific name of *bambusa blumeana* and says that is the best bamboo for making instruments. Vic recommends this species because it possesses the right thickness, ideal fiber density, and a bright sound, making it easy to tune properly. Initially, the Toledos sourced their *bambusa blumeana* from their hometown in Nueva Ecija, where right behind their old house bamboos grew abundantly, but for a time, they were also sourcing bamboos from Montalban.

Bamboo, however, is not the sole material for the Toledo instruments. *Narra* wood, for example, was used for their xylophone slabs, particularly the marimbas. Chino explains that slabs made from *narra* wood give a good sound and at the same time, are sturdy and do not break so easily. According to him, they also tried *yaka*, but it had no sound at all. Yet bamboo is still a necessary material for the marimbas, as it is used as resonators to amplify the sound of the slabs. A quick glance shows these bamboos to be of uneven length. The main reason for this is that the nodes, and not the length or entirety of the bamboo, are the ones tuned to the slabs. This is why, according to Chino, bamboo tubes are cut from node to node. In fact, when bamboo tubes don’t have a

node – as seen in some of the bamboo resonators – their bottoms are sealed off with an artificial cover to simulate a node so that they can be tuned to the necessary pitch. For some of the lower pitches, tuning the tube is not as important, but the tube is still necessary for increasing volume. As for the other resonators, some are there just for aesthetic purposes, in order to fill in the gaps in between the actual bamboo resonators.

When asked about what the main innovation contributed by his father and grandfather could be, Chino said that it is not in the instrument design, but rather in the process of making the instrument. Truly, the meticulous process is astounding in itself and is the main reason for their quality and durability. As his son Chino would say, Vic Toledo's efforts in perfecting bamboo instrument making and design stem from his passion in treating the bamboo not as a toy or souvenir but an instrument built to last a lifetime.

Excerpt from the BMI Ethnography of Laguna's Biñan Kawayan Music Ensemble
"Ret. Col. Gilbert Ramos, Biñan Kawayan Music Ensemble, and Musikawayan"
 By Roan May D.G. Opiso

BAMBOO MUSICAL INSTRUMENTS

Mr. Ramos and his family made the following bamboo instruments that are used by the different ensembles that he leads.

INSTRUMENT	METHOD OF SOUND PRODUCTION	MELODIC RANGE	ROLE IN THE ENSEMBLE	FUNCTION IN RITUAL OR SOCIAL/CULTURAL ACTIVITY
<i>Angklung</i>	Sound is produced by shaking the instrument	1 pitch per instrument	Plays melodic line; fill in the chords	An ensemble, the instruments are played during fiestas and other celebrations, mostly functioning as entertainment music.
<i>Piangklung</i>	Sound is produced by shaking the instrument	Around 2 octaves	Plays lead melodic line	
<i>Bumbong</i>	Sound is produced by blowing air into the hole of the instrument	1 pitch per instrument/ bamboo tube	Functions Like a horn and tuba in lower pitch	
<i>Rambong</i>	Opening of bamboo tube is struck using rubber mallets	One and a half octave	Functions like a bass	
<i>Marimba</i>	Wooden keys are struck using mallets	2 octaves	Plays lead melodic line and counter melody	
<i>gabbang</i>	Wooden blades are struck using mallets	2 octaves	Plays melodic line and effective for marching	
<i>Pito</i>	Sound is produced by blowing air into the hole of the instrument	2 octaves	Plays lead melodic line and counter melody	
<i>Bamban</i>	Sound is produced by humming (like a kazoo)	Conforms the range of the voice of the player	Plays lead melodic line	

Angklung is a bamboo shaker which is an instrument that is tuned to different pitches (Angklung 2013). One person plays one angklung which corresponds to one note.

Piangklung is a set of angklung hanging on a wooden frame. It is arranged like a piano keyboard (Piangklung 2013) and is played by one person. This instrument plays the melody.

Bumbong is a blown bamboo tube instrument positioned transversely from the player's mouth. In the Biñan Kawayan Music Ensemble (BKME), it is grouped chromatically in twos or threes and functions like a trombone and tuba in a marching band.

Rambong is a set of 13 bamboo tubes positioned vertically in a frame. The player hits the opening of the bamboo tubes using a rubber mallet. It is arranged chromatically like the marimba and has a range of one and a half octave. It is the bass of the ensemble and functions like a tuba. Design-wise, the marimba used by BKME is similar to the western classical marimba. However, its keys are made of narra while the resonators are made of bamboo. It has a range of two octaves.

Gabbang is a bamboo xylophone, placed in a wooden box.

Pito is a transverse bamboo flute which came from the Igorot. It is the main melodic instrument of Musikawayan and has a range of two octaves. In the original Musikong Bumbong, there were four players playing this instrument. More pito players were added in the next generations of the band.

Bamban, a bamboo instrument that is similar to the kazoo, was invented by Mr. Ramos' great-grandfather. The actual instrument is made of buho (a species of bamboo), while the membrane is made from a piece shaved from the inner part of fresh kawayan. However, when the membrane gets wet, it can no longer be used and must be changed. From the original bamban developed by their ancestors, the Ramos family further developed the bamban and used cellophane to attach the membrane to produce a better sound and longevity.

Musikawayan also has a bamboo clarinet with a modern mouthpiece. Before, they used a reed and a mouthpiece made of bamboo, but when they tried to use a plastic mouthpiece from an alto saxophone, they found it easier to use.

Excerpt from the BMI Ethnography of Sta. Mesa's Banda Kawayan

"Concertized Bamboo Music Cultures"

By Patricia Marie U. Dizon

In terms of materials used in Philippine instrument-making cultures, bamboo is one of the commonly used. While the *bumbong* is often associated with the *musikong bumbong* bands in Bulacan, Calabig uses other types of bamboo. Apart from the *bumbong*, he often uses Chinese bamboo as he finds it more durable than the other varieties of bamboo. Known as *kawayang Tsina* in Paete, Laguna, this variety is also more malleable but less brittle when dry. *Buho*, a thinner variety of bamboo, is used too, albeit sparingly. Other varieties of bamboo that have thorns may be used in house construction, but not in instrument-making, as they are difficult to shape and become brittle when dried. Wood is used, with mahogany and acacia bearing the best sound.

The ensemble's instruments are the marimba, *angklung*, *panpipes*, *bumbong*, *kalatok*, *kalagong*, and *kiskis*.

Some of the instruments used by Banda Kawayan are quite familiar to the audiences. The marimba, for example, is one of the generally used musical instruments in the world, but while the standard marimba seemed bulky for ingress to and egress from the performance venue, Fred's version is more portable and foldable. "*Makikita ninyo yung k'wan, dapat collapsible pagkatapos dapat magaan pero yung durability hindi. Magagaan ang kahoy kong ginagamit, lalo na ngayon makikita mo yung nakasandal na 'yon,*" he said. The portability of the bamboo marimba is something unique to the Banda Kawayan's set of musical instruments. Another distinct feature of Fred's version of the marimba is that the resonators are made of bamboo tubes, instead of the usual steel.

Apart from the marimba, some of Banda Kawayan's musical instruments which are commonly known are the *angklung* and the panpipes. The *angklung* is an Indonesian musical instrument which is played by shaking the lower part while supporting the upper part. Meanwhile, the panpipes are a bunch of small bamboo pieces put together and are played by blowing. In the Banda Kawayan, the *angklung* section plays the chords, while the panpipe section plays the ensemble's counterpoint. In terms of appearance and construction, while Banda Kawayan's version of the *angklung* looked almost the same as its Indonesian counterpart, it was a different case for the panpipes: Normally, panpipes are bound equally on top with varying sizes only towards the bottom part. Banda Kawayan's version, on the other hand, has a neat arrangement of the pipes at the bottom, but its front and back parts are bound unevenly. During the 1970s up to the 1980s, Fred made an attempt to have his *angklung* design patented. However, he did not renew it anymore, as registration for patented designs is not only costly but also tedious, as it had to be returned when the design did not meet the government's standards.

The rest of Banda Kawayan's musical instruments are rather distinct, which allowed both Fred and the ensemble to make their mark in the instrumental ensemble performance scene here in the Philippines and abroad.

First is the *bumbong*. Although similar to the commonly designed panpipe in shape (a set of hollow bamboo pieces) and manner of playing (by blowing), the *bumbong* has holes on the surface, as the instrument is meant to be blown on such portion. In an ensemble, it is divided into two sections: the "bass" section, which plays the pieces' lower wind parts and the "horn" section, which plays the higher wind parts.

Another distinct instrument of the ensemble is the *kalagong*. It has the same frame as the marimba, but the bamboo poles are hung inside the frame. Hence, the name “*kalagong*” is a portmanteau between “*kala*” (Calabig) and “gong,” which is the instrument maker’s brainchild. The *kalagong* is the ensemble’s bass parts.

Another instrument named after Fred is the *kalatok*, which is also a distinct musical instrument of Banda Kawayan. It is played similarly to the Latin American clave and the Kalinga *pateteg* by being struck by a stick. While the clave is a wooden block, and the *pateteg* a bamboo blade, the *kalatok* is a hollow bamboo tube with a resonator found on the surface and closed ends.

The *kiskis* is a scraper also made of Chinese bamboo. It is tuned through holes in between slits. Scrapers have been existent in other music cultures, especially Afro-Latin. Both the Afro-Latin percussive scrapers, known as the *guiro*, and Banda Kawayan’s *kiskis* are played by being scraped by another stick. However, while the *guiro* is gourd-shaped, the *kiskis* is rather elongated.

The *kalatina*, another instrument made by Fred, is a hung quill-shaped musical instrument with a sound hole at the center made of *buho*. It is played by being struck.

A pitched wooden block musical instrument made of Chinese bamboo, which still had no name, was used by Calabig during the group’s tour in Spain. It is played by striking such wooden block with a mallet.

Fred also makes wind musical instruments, such as flutes. While he sometimes creates Cordillera musical instruments, such as the *kubing* and *tongatong*, it is not as lucrative as the making of Banda Kawayan’s musical instruments previously stated. He also prefers to make solid-sounding musical instruments, which are able to resonate even with the absence of microphones onstage.

Some of the instruments are old, the *bumbong* being the oldest at twenty years. They are carefully handled, but in case of breakage, the instruments are wrapped with a tape, which do not affect the instruments’ sounds. Sometimes, they are repaired using Mighty Bond. However, when instruments are already difficult to repair, they are replaced.

The instruments are tuned at 440 Hz. Initially, they used a pitch pipe. However, due to technological advancements, they now use a chromatic tuner, which is widely available through Google Play and App Store downloads. One of the more modern tuner’s advantages is that a visual representation of such pitch is now shown on-screen, either on a tablet or on a smartphone.

Banda Kawayan’s musical instruments are stored in its headquarters inside the Communication Foundation for Asia (CFA) compound, which is owned by the Catholic Church. The group was able to acquire such place when they performed during the opening of the CFA compound in the 1970s. As the group performs during tours, Calabig always sees to it that the instruments are collapsible and fit inside the balikbayan box (30x20x10). He also ensures that the instruments would not exceed the maximum weight required for hand carried luggage. The group’s costumes are designed by renowned

fashion designer, Renee Salud, who is the cousin of Fred's sister-in-law. Fred stated that Salud does not charge the group an exorbitant amount of money. "*Kilalang-kilala ko si Mama Renee so, pakisamahan lang,*" he said.

Fred finds the musical instruments useful in classroom settings, especially the *angklung* and *tongatong*. Based on his experience, anyone who has both the reading and writing ability will be able to play such instruments. Having a concept of good musical timing is also essential.

Fred was able to pass on the bamboo musicmaking activity to his family, particularly to his firstborn child and second child. However, they were not able to adapt the instrument-making process, as it was too challenging, and it would require skills in carpentry and use of sharp tools (e.g., saw, *katam*). At present, his townmates, Jun Cadapan (a former Paete municipality employee) and Rodrigo Cadapan (a former trumpet player of the Philippine National Police Band), who are also skilled woodcrafters from Paete, Laguna, are the apprentices who help him in instrument-making. He is currently making a living by creating musical instruments for Far Eastern University and another one for Tarlac and Canada. Recently, the Nueva Vizcaya chapter of the United Methodist Church purchased a twenty-five-piece set of *angklung* from him, which costed Php 1,200 per piece. Such set contained chromatic pitches. According to Fred, he earns more money in instrument-making.

Excerpt from the BMI Ethnography of Dulag's Karatong Festival

"Dulag, Leyte: Presenting Identity through Bamboo Musical Instruments"

By Roan May D.G. Opiso

KARATONG FESTIVAL

Karatong Festival is the annual *fiesta* of Dulag, Leyte. It highlights the Waray bamboo percussion karatong, which in ancient times was used by the Dulagnons as an alarm when an impending danger or calamity would threaten their town (Cagara, interview, 19 October 2019).

Mr. Orlando Cagara, Secretary to the Sanggunian Bayan of Dulag, shared that Dulag's previous festivals used to be heavily patterned after the Sinulog of the neighboring province of Cebu. In 2019, however, the Dulag local government unit (LGU), under Honorable Mayor Mildred Joy Que, decided to impart their own identity in their festival. At the same time, they wanted an all-bamboo music featured in their festivity. Hence, they elected to develop the *karatong* and other bamboo musical instruments.

Why Bamboo?

Upon developing the new concept for their festival, the Dulag LGU looked into the resources of their land. Dulag, according to Cagara, is generally a flat land, and though they do not have a forest, the town has six to seven species of bamboo. They have pattong, tangnan, kawayan, bagakay, among others.

BAMBOO INSTRUMENTS

In preparation for the Karatong Festival, the Dulag LGU invited Prof. Jocelyn Guadalupe of the UP College of Music and Philippine Society of Music Educators and Mr. David Dino Guadalupe of the UP Center for Ethnomusicology to give a workshop on bamboo music. This workshop was attended by teachers of the different participating schools in the municipality and was later followed by rehearsals and further encouragement from the local government.

During the said workshop and the succeeding preparations for the festival, the schools were able to create different types of bamboo instruments. Additionally, the bamboo instrument makers named their creations based on their Leyte/Samar heritage, congruent with their aim of highlighting the Dulag identity in their festivities. Hence, the following instruments came to being.

Halo or Bayawak

The *halo or bayawak* is a set of three (3) long bamboo tubes laid in a row. Its name came from the Waray word halo, meaning *bayawak* in Filipino or monitor lizard in English, because its length is similar to that of the said animal.

The instrument is made from the *bolo* bamboo, which is locally known as *kayali*. The instrument is played by striking the holes at the end of the bamboo using a rubber mallet. (Some players improvised using a rubber slipper). Each bamboo tube of the *halo* has its own note. The longer the length of tube, the lower the pitch. At the same time, the shorter the tube, the higher the pitch.

The *halo* has different sizes and each one is tuned to itself; meaning, a *halo* is not necessarily tuned the same way as other *halos*. Played in an ensemble, its function is comparable to that of the tenor drum of a marching band.

In creating the *halo*, Romeo and Paul of San Jose National High School first took the measurements of a bamboo pole. They then hollowed out the inside, and used a pitch pipe to get the pitch of each tube. Paul started with the lower 'do' before going up to the highest note. Mr. Go, another teacher from their school, made the decorative design of the instrument.

Karatong

As part of an ensemble, the *karatong* has a function similar to that of the snare drum of a marching band. Though it has been a traditional instrument in Dulag, the instrument makers in the workshop made innovations on its design to suit their needs. For example, they came up with a *karatong* version which has a handle on one end so that it can be held by one hand and beaten with a bamboo stick by the other hand. This style makes it convenient for the drummer to dance and to play the instrument at the same time. Another style of the *karatong* has the instrument placed on a stand, which is then strapped to the player. This set-up allows the instrumentalist to simultaneously dance and play the instruments using two sticks. A third version is a set of *karatong* grouped in three's and placed in a frame.

Like the functional innovations made on the instrument, the decorative design of the *karatong* also varies depending on the maker. In San Jose, for example, some of their *karatong* were styled like a fish.

Karatoktok

The Dulagnons who attended the bamboo music instrument seminar also made a bamboo lyre or xylophone which they called *karatoktok*. It is played using a bamboo stick as a mallet. Like the *halo*, the *karatoktok* is also made from *kayali*.

The *karatoktok* in San Jose NHS was tuned by Paul and used the notes of the song "Si Filemon" which they performed in the festival.

Other Instruments

Aside from the *halo*, *karatong*, and *karatoktok*, the instrument makers also made other different bamboo instruments. They have another bamboo xylophone similar to a *gabbang*, and a huge bamboo clapper. The bamboo clapper though was used more as a prop than an instrument.

They also have a smaller version of *halo* called by Romeo as *munting sawa*. These shorter bamboo tubes are grouped together (one version has at least 8 tubes) in one row and set vertically in a frame, unlike the *halo* which is laid down horizontally. The *munting sawa* was acquired after the *Karatong* Festival and is reserved to be played in future activities.

Instrument	Method of Sound Production	Melodic Range	Role in the Ensemble	Function in Ritual or Social/Cultural Activity
<i>Halo</i> (or <i>bayawak</i>)	Hole at the end of the bamboo tube is hit with a rubber mallet	Range depends on the maker (ex. length of the bamboo tube)		Part of an ensemble, the instruments are played during fiestas and other celebrations, mostly functioning as entertainment music; accompaniment to street dancing.
<i>Karatong</i>	Body of bamboo is hit with a stick		Functions like a “snare drum”	Part of an ensemble, the instruments are played during fiestas and other celebrations, mostly functioning as entertainment music; accompaniment to street dancing. Traditionally used as a warning device.
<i>Karatoktok</i>	Bamboo slabs are hit with a bamboo stick	Instrument tuned to itself (range may vary depending	Plays melody	Part of an ensemble, the instruments are played during fiestas and other celebrations, mostly functioning as entertainment music; accompaniment to street dancing.
Bamboo xylophone (<i>gabbang</i>)	Bamboo slabs are hit with a stick	(No data)	(No data)	(No data)
<i>Munting sawa</i>	Hole at the end of the bamboo tube is hit with a rubber mallet	(No data)	(No data)	(No data)

Excerpt from the BMI Ethnography of Angono's Musikong Bumbong

"Musikong Bumbong: Musika Kawayan of Angono, Rizal"

By Juliet R. Bien

DESCRIPTION OF THE INSTRUMENTS

A. French horn

Simulating a brass French horn, the *musikong bumbong* prototype has the following measurements: length 20 inches, width 17 inches, height 3 inches, diameter of biggest barrel 2 inches, and valve 5 inches. It has three valve levers interconnected side by side by a 7-inch bamboo dowel. The "fake" mechanism rests on a lead pipe which coils around the body of the instrument spanning 51 centimeters. From the valves, the coiled pipe passes through a 3-inch bamboo connector with an approximate diameter of 1 inch. The pipe is then attached to the bigger vertical pipe (17 inches length and 3 inches diameter) which is the opening of the instrument. The air column spans the length of the mouthpiece traversing towards the main vertical chamber. The sound mechanism does not include the valves and the coiled pipes. The tone is produced through the transverse pipe towards the vertical chamber. The endpipe is slashed with 7-inch incisions to produce a buzzing sound. The endpipe functions as the bell valve. Unlike the brass French horn and the *musikong bumbong* instruments in other towns like Bulacan, the *bumbong* version of Angono's Musikang Kawayan does not have a bell.

B. Trombone

The trombone prototype is a composite of the main body and its removable slide. Like the other *bumbong* instruments, the bamboo design is a simulation of a brass trombone. The air is blown through the mouthpiece channeled towards the tip of the instrument, with the slide as decoration. The buzzer takes the place of the bell and the bell valve. The measurements are as follows: length (with unstretched slide) 42 inches, length of removable slide 24 inches, mounting rod for the removable slide 13 inches, total length of removable slide curved pipe 51 inches, removable slide brace 4 inches, length of main body 55 inches, incised buzzer 4 inches, main air chamber 19 inches, and airpipe diameter 1.75 inches.

The sound is produced by blowing the air through a detachable funnel-shaped mouthpiece carved from a piece of wood. The platform supports the main body of the instrument and holds the parallel rods from which the removable slide is mounted. The air column chamber traverses down to the left and curves toward the bell cylinder. The sound passes through a series of bamboo pipes with various lengths: a medium-sized vertical 7-inch chamber, a small U-shaped 8-inch pipe, 0.5 inch connected to a 2-inch medium sized joints, and a 28-inch barrel with a 2-inch diameter.

The removable slide consists of an elongated deep and narrow U-shaped bamboo pipe spanning 55 inches supported by two braces on both ends. The braces consist of a medium-sized bamboo cylinder and a smaller pipe which are located closer to the mouthpiece and towards the end of the instrument, respectively. This mechanism is randomly slid and does not produce musical sounds. During performance, the slide is moved randomly as an imitation of how trombones are played in the brass band.

C. Trumpet

The trumpet is a singular piece of instrument with the following measurements: length 22 inches, width 7 inches, height approximately 5 inches, total length when uncoiled 56 inches, thickest diameter 1.25 inches. From the mouthpiece, the lead pipe loops into a curved body which consists of three interconnected bamboo pipes imitating the outline of a brass trumpet. The three parallel 7-inch valves strung together by a bamboo rod is mounted into the loop by two 5-inch braces on the left and right sides with a distance of two inches from the valves.

A narrow bamboo pipe is curved connecting the straight barrels towards the outlet. The largest pipe with a diameter of 1.25 inches takes the place of the bell. Like the other instruments of the set, the endpipe is a cylindrical piece of bamboo and incised towards the end to produce a buzzing sound. Air is simply blown into the mouthpiece and adjusted by the musician's lips according to the repertoire. The valves are randomly played to imitate trumpet playing in the Western brass band tradition.

D. Euphonium

The largest instrument in the collection, the euphonium consists of two vertical pipes supported by two horizontal braces. The three valves constructed parallel to each other are connected by narrow pipes looping its way towards the lead pipe simulating the design of a brass euphonium. The measurements are as follows: longer vertical pipe right side 29 inches, horizontal beams 13 inches, shorter vertical pipe 14 inches, width 12 inches, length of coiled frame 50 inches, valves 7 inches, finger stop diameter 0.25 inches, and endpipe diameter 3 inches. The euphonium is the only instrument with a finger stop bored through the lower corner of the larger vertical pipe at the opposite side from its player.

The simulation of its brass counterpart is done by constructing the coiled mechanism on top of the larger pipes functioning as the main frame supporting the valves and the coils. From the mouthpiece, the sound is produced passing through the horizontal beam, then through the curved pipe at its end as it connects to the endpipe. Like the rest of the instruments, the bell is replaced by the endpipe incised with 7-inch slits and produces a buzzing sound. The tones are refined by adjusting the finger stop. The rest of the coiled mechanism and the valves are not functional. Like the rest of the instruments, they are designed to imitate the brass euphonium but do not produce any sound.

The instrument's "aesthetic look" consists of the valves and the coils mounted on the bigger pipes. Between the first and the second valve, a narrow bamboo pipe is curved to form a U-shaped pipe towards the body of the shorter vertical pipe. A bigger U-frame enclosing the valves is constructed approximately two inches on the left side of the first valve whose end is attached to a bigger vertical pipe attached to the smaller vertical supporting pipe frame. A curved enclosure is completed by a U-shaped narrow pipe connected to the lead pipe.

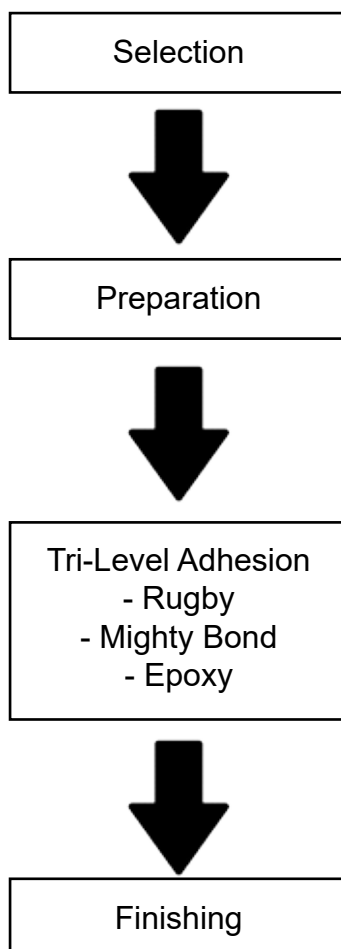
Excerpt from the BMI Ethnography of Bulacan's Musikong Bumbong

"Mr. Rodrigo Anastacio and the Banda Santa Clara Musikong Bumbong of Obando, Bulacan"

By Daniel Ray C. Eramis

PRODUCING THE BAMBOO MUSICAL INSTRUMENTS

Based on the discussion with Sir Rod, the diagram will explain the analyzed systematic process of producing the bamboo musical instruments from selection of materials to the final form of an instrument:



For Sir Rod, the best time to choose bamboo for an instrument is when it is fresh, so as it is easier to cut. He prefers to source his supply from Bulacan, usually from the areas of Plaridel, Pulilan, and Baliwag as it is available all year round within the province. He chooses *buho* (a hollow kind of bamboo) as small tube pieces for regular to small sized instruments and large bamboo species (size of saucer in diameter) for the bigger sized instruments like the bajo (tuba/bass).

After selecting, the specific parts of the bamboo are cut to fit the need of an instrument. They are then prepared by sanding out unnecessary parts and treating the insides with varnish to avoid weevil infestation.

Soon after the treatment had dried up, the bamboo parts are structured together and undergo three layers of adhesives:

- 1) Initially, the parts – inclusive of auxiliary parts – are joined with Rugby.
- 2) Once dried, the structure is reinforced with Mighty Bond. Sometimes small bamboo sheets or sticks are used to strengthen specific spots.
- 3) The joints are covered with Epoxy to fully strengthen the structure.

After the Epoxy has dried up, the excess is sanded off to provide aesthetic look for the finished instrument. Sometimes, if deemed necessary, certain areas are given another layer of adhesive. Once done, the whole instrument is protected with a layer of varnish.

For instrument making process and design patent concerns, the table below describes the innovative technique/s that Sir Rod had applied in his craft:

Instrument		General <i>Bumbong</i> Instruments	Folk Obando – Banda Santa Clara	Individual Sir Rod
Pitched	Flute (Recorder)	Made of bamboo	-	-
	Saxophone	Originally, all instruments were made of straight bamboo cuts, having varying sizes and lengths depending on the desired role.	Took shape with the use of carabao horn as reed, and tin can as resonator	The most unique methods that Sir Rod utilized in producing the instruments are as follows: 1) Use of varnish to protect instruments from infestation. 2) Use of the trilevel adhesion method (rugby, mighty bond, and epoxy). 3) Use of rattan for the bells of specific instruments using them.
Buzz	Flute		Instruments began taking shape similar to their brass band counterparts with use of kazoo as primary producer of sound	
	Transverse Flute			
	Trumpet			
	Clarinet			
	Baritone			
Trombone				
Single-Pitched	French Horn		Like all instruments in the ensemble, took shape into their brass band counterparts with the spit type	
	Bajo (Tuba)			

Aside from the innovations presented in the table above, some notable features of the bamboo musical instruments of Banda Santa Clara are the keys attached in the instruments like those in French Horns which do not really have any function but as accessories for performative purposes. They are made of button-shaped plastic attached to spring mechanisms to make them realistically look like their counterparts.

THE INSTRUMENTS AND THEIR ROLES

The ensemble of Banda Santa Clara Musikong Bumbong can be categorized under three methods of sound production: 1) Pitched Instruments, 2) Spit-type (like in brass instruments), and 3) *Kazoo*-based or Buzz Membrane. Due to the limitations of majority of the instruments, they play songs that are usually in or are transposed to the key of C Major. The table below helps to explain the methods of sound production, range, and role of each instrument in the ensemble.

Instrument		Method of Sound Production	Melodic Range	Role in the Ensemble
Pitched	Flute (Recorder)	- A kiss-flute method where a reed is fixed on the instrument - Pitch is manipulated by covering holes	c1 to c3 (2 octaves) but can be higher if advanced	Lead Voice
	Saxophone			
Buzz Membrane	Trumpet	- Using the carabao horn as reed and tin can as resonator - Pitch of the sound produced is manipulated by covering holes.	Range depends on the performer. Instrument can make clear sound from a to a2	Lead Voice
	Flute & Transverse Flute			Lead / 2nd Voice
	Clarinet			Lead / 2nd Voice 2nd Voice
	Baritone			2nd / 3rd Voice
	Trombone			2nd / 3rd Voice
Single-Pitched	French Horn	Similar to brass instruments, The pressure of air affects resultant pitch	Between G to A (Overtone Possible)	Ostinato Response to Bass
	Bajo (Tuba)		Between C to D (Overtone Possible)	Down Beat Keeper
Percussion	Cymbals	Clashing (Metal)	-	For Emphasis Parts
	Snare Drum	Membrane and Rim Striking	-	Time Keeper, Signals the Band
	Bass Drum	Membrane Striking	-	

Excerpt from the BMI Ethnography of Maasin's Tultugan

"Tultug: The Root of Maasin's Tultugan Festival"

By Hermie F. Cartagena

INTRODUCTION AND HISTORICAL BACKGROUND

Bamboo plants are abundant in Maasin. One of the varieties that can be found in the area is the "kawayan tinik" and is produced yearly with more than 2 million poles. Thus, bamboo is the primary material in producing local products and handicrafts in Maasin. The place has been supplying bamboo poles and various handicrafts in Iloilo and has even been exporting them internationally (Marin, 2017). Being one of the primary exporters of bamboo products in the region and the country, this municipality in the central part of Iloilo became known as the bamboo capital of the Visayas.

In celebration of the abundant supply of bamboo products in the municipality, the Tultugan Festival was established. The festival, initiated or institutionalized by Mayor Mariano Malones in 1999, is considered a contemporary festival sponsored by the government (Muyco, 2016). Tultugan came from the word "*tultug*," which means the act of making sound by tapping or striking the bamboo instrument (Marin, 2017).

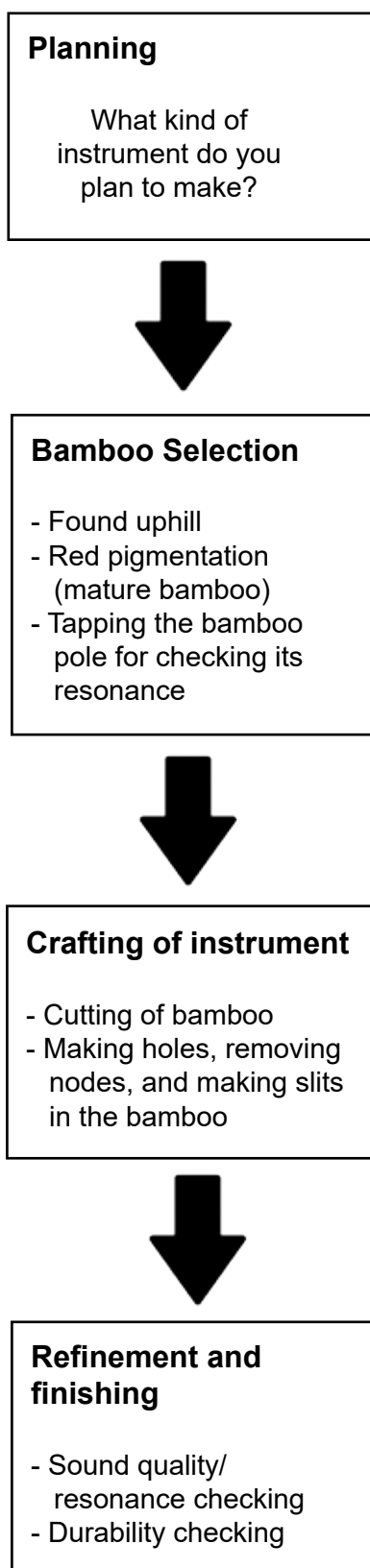
The Tultugan Festival is primarily dedicated to bamboo and its various uses and functions. Held every fourth week of December, it features street dancing and a tribal dance competition. The highlight of the celebration is the tribal dance competition that showcases the various usage of bamboo in the community. The performers show how to creatively utilize the bamboo materials as part of the performance, incorporating them into their props, costumes, and instruments. Aside from the street dancing and tribal dance competition, the festival also highlights the "*Gwapo Karabaw*," a contest wherein the participants adorn their water buffalos with various ornaments; "*Parada sang Litson*," where roasted pigs are presented in the streets through a parade; "*Rara Amakan*" contest, wherein the speed in weaving bamboos is tested; "*Kadang Race*," a test of balance; and the search for Miss Tultugan ("*News Today*" 2005, p. 1, paragraph 3).

The festival is a way of preserving the local culture for the next generations to come, and various approaches were applied in redesigning and modernizing the bamboo material. Likewise, various rhythmic patterns were composed and developed by local musicians to provide a dynamic and vibrant sound in the performance (Marin, 2017; Jacinto, 2013).

PRODUCING THE BAMBOO MUSICAL INSTRUMENTS

All bamboo instruments and products in the municipality of Maasin are made from the kawayan tinik variety. Ms. Rosemarie Bayag explained that the best time for the harvesting of the Kawayan tinik is from September to January. At this time, the bamboo is mature enough to be crafted as a musical instrument. The diagram below shows the systematic process of making a bamboo instrument from planning up to its final form.

For Bayag, the first step in making a bamboo instrument is planning. She explained that an instrument maker should have in mind what kind of instrument he or she desires to make.



Secondly, the selection of bamboo must be fit to be crafted as an instrument. For Bayag, the best bamboo material are those bamboos that are situated uphill and have red pigmentations already. She also taps the bamboo pole using a bolo to check for sound quality. For her, not all bamboo has the potential to “sing.”

Bayag further explained the science behind this process. According to her, bamboos that are situated uphill have less moisture in their fibers when compared to those bamboos found in the lowland area. She also added that bamboos found uphill are used to dry or humid climates. With that, the bamboo fibers are denser and sturdier. Also, she advised to look for the red pigmentation of the bamboo, which signifies that it is already mature enough and ideal to be constructed as an instrument. She further explained that mature bamboos with denser fibers are better because they produce a brighter and better sound resonance or what she called “*nagakanta nga instrumento*” (singing instrument).

The third step is choosing the appropriate part of the bamboo fit to be an instrument. According to Bayag, she only cuts and uses the *pusog* and the *puno* part of the bamboo – found in the lower part of the bamboo stem. Such parts have thicker bamboo flesh compared to its other parts. This is followed by carefully and neatly cutting or removing some bamboo nodes (for pipes or pitched instruments) and making a slit on one side of the bamboo (for rhythmic types of instrument like *tultug*). Bayag makes sure that this step is meticulously done for it may affect the sound quality of the instrument.

Lastly, refinement of the instrument is done by checking the sound quality (resonance) and durability. Bayag explained that once the bamboo instrument got cracked or damaged during the refinement stage, it is simply replaced. For her, the sound quality of the bamboo will be affected if adhesive will be applied to fix the broken part of the bamboo.

THE INSTRUMENTS AND THEIR ROLES

The instruments in the municipality of Maasin can be classified into two – rhythmic and pitched. Rhythmic instruments produce sound by striking, stamping, and clashing bamboos. Pitched instruments, on the other hand, produce melodies by striking the surface or tubes of the bamboo. These instruments are set in three octaves tuned to the key of C. The table below explains the method of sound production, melodic range, role in the ensemble, and function in social/cultural activity.

Instrument	Method of Sound Production	Melodic Range	Role in the Ensemble	Function in Social/Cultural Activity
Tultug	Patik (striking) of the stick on the tultug		Pulse/ timekeeper	Although the instruments may vary in sizes, their function can be taken as one – an accompaniment to Tultugan Festival. As an ensemble, the instruments provide a festive atmosphere in the festival highlighting the flexibility of bamboo as an instrument.
Bayong	Stamping (rubbing coating at the bottom part of the bamboo)		Bass	
Tala-tala	A hand-held instrument, the sound produced by clashing the bamboo		Clapper/ cymbals	
Paraspas	Hand-held instrument; Clashing of the bamboo strips		Rattle/ clapper	
Gabbang (bamboo xylophone)	Striking the surface of the bamboo	3 octaves tuned in the key of C	Melody	
Bamboo Pipes	Graduated hallowed bamboos. Striking the bamboo tubes from the top covering the holes.	3 octaves tuned in the key of C	Additional texture; melody	
Mini Pipes	Graduated hallowed bamboos. From the top, a rubber is stamped to the bamboo covering the entire bamboo circumference. The sound comes out from the bottom. However, the instrument can be reversed for different sound quality. Rubber stamping is still done at the top of the bamboo instrument covering the entire hole.	3 octaves tuned in the key of C	Melody	

Appendix B – Photos

BAMBOO ORGAN OF LAS PIÑAS



BAMBOO UKULELE OF CEBU



HuniUkulele's "bamboolele" (soprano size, teardrop shape)



HuniUkulele's "bamboolele" (concert size, standard shape)



HuniUkulele's "bamboolele" (tenor size, round shape)

HIMIG KAWAYAN OF QUEZON CITY, LAGUNA, STA. MESA

Pangkat Kawayan of Quezon City



Alugtog (Angklung)



Bumbong



Tipanklung



Tunog Hagdan



Chord Rattle



Marimba



Sungkit

Banda Kawayan of Sta. Mesa Manila



Angklung



Bumbong



Bumbong



Panpipe



Kalagong



Marimba



Kalatok



Kiskis

Biñan Kawayan Music Ensemble / Musikawayan of Laguna



Bumbong



Angklung



Tipanklung



Marimba

KARATONG FESTIVAL OF DULAG, LEYTE



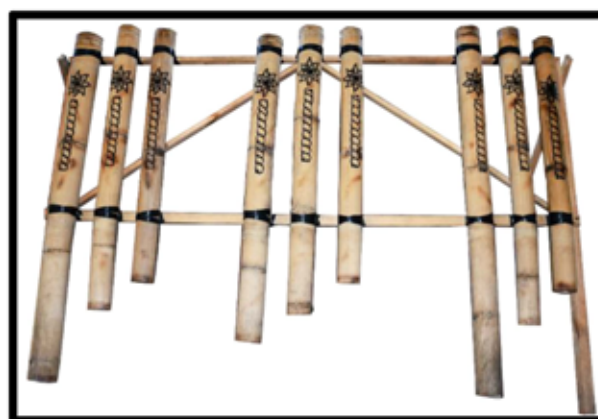
Karatong



Clapper



Karatoktok



Halo



Bayawak/Hagubhob

MUSIKONG BUMBONG OF ANGONO, RIZAL



Trumpet



Trombone



French Horn



Euphonium

MUSIKONG BUMBONG OF BULACAN

Clarinet



Trumpet



Transverse Flute



Trombone



Euphonium



Baritone



Bajo - Tuba



TULTUGAN MUSIC OF ILOILO



Gwapo Karabaw*



Kadang Race*



Tribal Dance Participant*



Tribal Dance Participant*



Tribal Dance Participant*



Musical Instruments of a Tribal Group*



Tultugan Instrumental Ensemble**



Street Parade Featuring the Paraspas**

*Photos of past Tultugan Festivals courtesy of Mr. Gilbert Marin ("My Beautiful Iloilo" blog, 2017)

**Photos from the 2019 Tultugan Festival, courtesy of Ms. Rosemarie Bayag

Appendix C – URL Links

Suggested YT URL links during the time of publication only

Description	Link
Module 1: Bamboo Organ of Las Piñas	
BMI video of the Bamboo Organ demonstration by Armando Salarza	http://tiny.cc/BMI-Bamboo-Organ
Module 2: Bamboo Ukulele of Cebu	
HUNI Soprano Ukulele	http://tiny.cc/YT-HuniUkuleles-Soprano
HUNI Tenor Ukulele	http://tiny.cc/YT-HuniUkuleles-Tenor
Module 3: Himig Kawayan of Quezon City, Laguna, Sta. Mesa	
Angklung – Himig Kawayan video tutorial	http://tiny.cc/YT-Angklung1
Indonesian Angklung	http://tiny.cc/YT-Angklung2
Angklung Performance	http://tiny.cc/YT-Angklung3
Marimba – Himig Kawayan video tutorial	http://tiny.cc/YT-Marimba1
Philippine Music: Dipolog Marimba Nais Ko	http://tiny.cc/YT-Marimba2
Banda Kawayan Plays Pinoy Music	http://tiny.cc/YT-Marimba3
Himig Kawayan, a video tutorial	http://tiny.cc/YT-Bumbong
Pangkat Kawayan	http://tiny.cc/YT-PangkatKawayan1 http://tiny.cc/YT-PangkatKawayan2 http://tiny.cc/YT-PangkatKawayan3
Musikawayan /D' Original Musikong Bumbong 1896	http://tiny.cc/YT-Musikawayan1 http://tiny.cc/YT-Musikawayan2 http://tiny.cc/YT-Musikawayan3 http://tiny.cc/YT-Musikawayan4
PUP Banda Kawayan/Banda Kawayan Pilipinas	http://tiny.cc/YT-BandaKawayan1 http://tiny.cc/YT-BandaKawayan2 http://tiny.cc/YT-BandaKawayan3
Bocaue Central School Musikawayan	http://tiny.cc/YT-Bocaue1
Musikawayan ng DepEd, Bocaue	http://tiny.cc/YT-Bocaue2
Piano Chords Tutorial	http://tiny.cc/YT-PianoChords
Basic Articulation	http://tiny.cc/YT-BasicArticulation
Module 4: Karatong Festival of Dulag, Leyte	
BMI documentation of Labiranan Festival of the 2019 Karatong Festival by San Jose National High School, Dulag Leyte	http://tiny.cc/BMI-Dulag-Karatong
BMI documentation of "Commitment to Peace" speech choir presentation by the Youth of Dulag	http://tiny.cc/BMI-Dulag-Choir

Module 5: Musikong Bumbong of Angono, Rizal	
BMI interview with Prof. Jose Rommel “Jopat” Gragera, conductor of Angono Wind Ensemble, and Angono Bamboo Instrument Maker	http://tiny.cc/BMI-Gragera-Interview
The Swan: Musikong Kawayan Bamboo Ensemble	http://tiny.cc/YT-TheSwan-MKBE
Pilipinas Got Talent Season 5 Auditions: Alicia Bohol Musika Kawayan Bamboo Musicians	http://tiny.cc/YT-PGT-AliciaBohol
Musiko 2013 #14 Angono Wind Ensemble	http://tiny.cc/YT-AngonoWindEnsemble
“Sa Ugoy ng Duyan” sung by Leah Salonga	http://tiny.cc/YT-UgoyNgDuyan
Module 6: Musikong Bumbong of Bulacan	
BMI interview with Rodrigo Anastacio, Bulacan Bamboo Instrument Maker	http://tiny.cc/BMI-Bulacan-Anastacio
BMI documentation of Marcelo H. Del Pilar students playing bumbong at Hiyas ng Bulacan 2020, Malolos, Bulacan	http://tiny.cc/BMI-Bulacan-MHDP
Sta Clara Pinung-pino - Fertility Ritual/Music & Dance of Bulacan - NAMCYA 2012 – CCP	http://tiny.cc/YT-StaClara-Pinungpino
Bamboo French Horn: Katrina Marzella on the Baritone: The Swan (Il Cigno) Noah 8th Grade French Horn Solo Ensemble	http://tiny.cc/YT-TheSwan-KMarzella http://tiny.cc/YT-FrenchHornSolo
Module 7: Tultugan Music of Iloilo	
YouTube videos on Tultugan Festival	http://tiny.cc/YT-Tultugan-Festival1 http://tiny.cc/YT-Tultugan-Festival2 http://tiny.cc/YT-Tultugan-Festival3 http://tiny.cc/YT-Tultugan-Festival4

Appendix D – Music Scores

Module	Title
Bamboo Ukulele of Cebu	“Tao, Tao po”
Himig Kawayan of Quezon City, Laguna, Sta. Mesa	“Manang Biday” “Pambansang Awit”
Karatong Festival of Dulag, Leyte	“Si Felimon”
Musikong Bumbong of Bulacan	“Sta. Clara Pinung-Pino”
Tultugan Music of Iloilo	“Tatlo ka Pato”
(Additional pieces)	“Bahay Kubo” “Balay sa Buntod”

Tao, Tao Po

Tagalog Folk

G D7 G



Ta-o, ta-o po, mayba - hay na ba - to, buk - san ang bin-ta - na't, ta-yo'y mag - pan-dang - go. Kung

9 C G D7 G



wa-lang gi - ta - ra, mas-ki na bi-la - o, ma - ki-ta ko la - mang ang da - la - ga nin - yo. Ang

17 D7 G



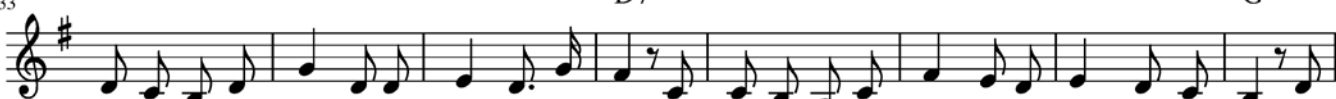
da-la-ga nin - yo a-yaw pa - li-ga - wan. Pu - ma-sok sa kwar - to't nag-sa - kit - sa-ki - tan. Ti -

25 C G D7 G



na-nong ng i - na kung a - nong da-hi - lan. Sa - got ng da-la - ga'y ma-sa - kit ang ti - yan. A -

33 D7 G



gad nag-pa - ta - wag da-la - wang me-di - ko. Pi - nag-tig - i - sa - han ang da - la - wangpul - so. Sa -

41 C G D7 G



bi ng me-di - ko wa-lang sa - kit i - to. Sin - ta ng pag-i - big u-mak - yat sa u - lo.

Transcribed by David Dino S. Guadalupe

Manang Biday

Isinaayos ni:
Genaro Cesar Edilo Jr.
10 Pebrero 2016

Medyo Mabilis

Score for the first system of "Manang Biday" (Tempo: Medyo Mabilis). The score is in 3/4 time and features four instruments: Gabbang Marimba, Angklung, Gitara, and Bumbong.

Gabbang Marimba: Treble clef, 3/4 time. The melody starts with a quarter rest, followed by eighth and sixteenth notes, and ends with a half note.

Angklung: Treble clef, 3/4 time. The accompaniment consists of chords and single notes, including a half note at the end.

Gitara: Treble clef, 3/4 time. The accompaniment features a steady eighth-note pattern with chords F, C, G7, and C.

Bumbong: Bass clef, 3/4 time. The accompaniment consists of a steady eighth-note pattern.

Score for the second system of "Manang Biday". The score is in 3/4 time and features four instruments: G M, A, G, and B.

G M: Treble clef, 3/4 time. The melody starts with a quarter rest, followed by eighth and sixteenth notes, and ends with a half note. Dynamic: *mf*.

A: Treble clef, 3/4 time. The melody consists of eighth and sixteenth notes. Dynamic: *mf*.

G: Treble clef, 3/4 time. The accompaniment features a steady eighth-note pattern with chords C, G, G7, C, F, C, F, and G. Dynamic: *mf*.

B: Bass clef, 3/4 time. The accompaniment consists of a steady eighth-note pattern. Dynamic: *mf*.

Score for the third system of "Manang Biday". The score is in 3/4 time and features four instruments: G M, A, G, and B.

G M: Treble clef, 3/4 time. The melody starts with a quarter rest, followed by eighth and sixteenth notes, and ends with a half note. Dynamic: *f*.

A: Treble clef, 3/4 time. The melody consists of eighth and sixteenth notes. Dynamic: *f*.

G: Treble clef, 3/4 time. The accompaniment features a steady eighth-note pattern with chords C, F, C, C, G, G7, and C. Dynamic: *f*.

B: Bass clef, 3/4 time. The accompaniment consists of a steady eighth-note pattern. Dynamic: *f*.

20

G M

A

G

B

F C G G7 C C C

26

G M

A

G

B

F C G G7 C G7

ff

ff

ff

ff

The musical score for 'The Rose Tree' is presented in five staves, labeled G, M, A, G, and B from top to bottom. The key signature is one sharp (F#) and the time signature is 3/4. The score begins with a double bar line and a repeat sign. The first staff (G) contains a melody with a first ending bracketed over the final two measures. The second staff (M) provides harmonic support with chords. The third staff (A) contains a melody with a first ending bracketed over the final two measures. The fourth staff (G) contains a melody with a first ending bracketed over the final two measures. The fifth staff (B) contains a bass line. The score concludes with a double bar line and a repeat sign.

The musical score for 'The Rose Tree' is presented in five staves, labeled G, M, A, G, and B. The score begins at measure 19 and continues through measure 27. The key signature is one sharp (F#). The staves are arranged vertically, with G at the top, followed by M, A, G, and B at the bottom. The notation includes various musical symbols such as treble and bass clefs, key signatures, and dynamic markings. The score is divided into measures by vertical bar lines. The G staff (top) features a treble clef and a key signature of one sharp. The M staff (second) features a treble clef and a key signature of one sharp. The A staff (third) features a treble clef and a key signature of one sharp. The G staff (fourth) features a treble clef and a key signature of one sharp. The B staff (bottom) features a bass clef and a key signature of one sharp. The score includes various musical notations such as notes, rests, and dynamic markings. The score is divided into measures by vertical bar lines. The G staff (top) features a treble clef and a key signature of one sharp. The M staff (second) features a treble clef and a key signature of one sharp. The A staff (third) features a treble clef and a key signature of one sharp. The G staff (fourth) features a treble clef and a key signature of one sharp. The B staff (bottom) features a bass clef and a key signature of one sharp. The score includes various musical notations such as notes, rests, and dynamic markings. The score is divided into measures by vertical bar lines.

74

Si Felimon

(Visayan)



Transcribed by David Dino S. Guadalupe

Santa Clara

Sayaw (dance) sa Obando

Intro

10
San-ta Cla - rang pi-nong pi - no ang pa - nga - ko ko ay ga - ni - to.

18
Pag-da - ting ko po sa O - ban - do ay mag - sa - sa-yaw ng pan - dang - go.

26
A - ba - ru - ray, a - ba - rin - ding. Ang pa - nga - ko'y tu - tu - pa -

30
rin. A - ba - ru - ray a - ba - rin - ding ang pa - nga - ko'y tu - tu - pa - rin

Transcribed by David Dino S. Guadalupe

Tatlo Ka Pato

(Three Ducks)

G D7

Tat - lo ka pa - to ang na - ki - ta ko. Ma -

3 D7 G

tam - bok ma - ni - wang wa - la ka - pa - re - ho

5 G D7

U - ga - ling ang i - sa i - kog nya nag - a - al - sa

7 D D7 G

Hum - lad ang pak - pak kag mag - ka - pa - ka - pa.

Dito sa suba sila pakadto
Langoy diri, langoy didto isa duha tatlo,
Ugaling ang isaikog niya nag-aalsa,
Humlad ang pakpak kag magkapakapa

Transcribed by David Dino S. Guadalupe

Bahay Kubo

Traditional
arr.for FPRDI
Bamboo Ensemble
by D&J Guadalupe

$\text{♩} = 54$

Bamboo Marimba

Bamboo Tube

Bamboo Sticks

F G C C G7 F C C

11

BB Mrm.

BB Tube

BB stks

G G C C G7 F C F G C

21

BB Mrm.

BB Tube

BB stks

C G7 F C C G G C C

31

BB Mrm.

BB Tube

BB stks

G7 F C F G G

39

BB Mrm.

BB Tube

BB stks

C C G7 F C C

45

BB Mrm.

BB Tube

BB stks

G G C C G7 G7

51

BB Mrm.

BB Tube

BB stks

C F G G C C

Balay sa Buntod

(Visayan children's song)

C Am F C F G C Am

May ba-lay ga-may sa bun - tod Kang ta-tay na-nay kad - tong ti-nu-kod

5 Dm G7 C Am Dm G7 C F

May ba-tang ga-may na - min - ta - na bo - u - tan si - ya nga ba - ta

9 C Am F C F G C Am

Ang i-yang na-nay gar - bu - sa Nag - la-kat - la-kat sa kal - sa-da

13 Dm G7 C Am Dm G7 C Repeat

Nag - si-naw si-naw ang pi - ta - ka pag an - do wa - la'y kwar - ta

transcribed by David Dino S. Guadalupe

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